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Linearly interpolating missing values in time series helps little for land cover classification using recurrent or attention networks

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ABSTRACT

Satellite time series data, widely used for land cover classification, often contain missing values due to cloud contamination, which can negatively affect classification. Numerous strategies have been developed to reconstruct the missing values to produce regular time series for machine learning classifiers, among which the compositing followed by the linear interpolation is most widely used. However, the classification improvement of linear interpolation for land cover classification has not been examined. Recently developed deep learning models such as long short term memory (LSTM) and Transformer allow such examination as they can classify time series with missing values. In this study, we compared the time series composites with missing values (without linear interpolation) and the linearly interpolated time series composites (without missing values) for land cover classification. About 18 thousand Harmonized Landsat Sentinel-2 (HLS) images acquired over Amur River Basin of China (890,308 km^2) in 2021 were composited to 14 16-day periods. Two time series composites were classified, i.e., (i) the 16-day composites without interpolation that have on average 15.35% 16-day periods with missing values and (ii) the linearly interpolated 16-day composites with no missing values. The classifications showed that (1) between classifications with and without linear interpolation there was $< 0.2\%$ overall accuracy differences for the bidirectional LSTM (Bi-LSTM) and $< 0.5\%$ for the Transformer both of which were smaller than model training randomness; and (2) the computation time can be saved using composites without linear interpolation. The findings suggested that it is unnecessary to use the time-consuming linear interpolation in Bi-LSTM and Transformer-based land cover classifications. The findings were confirmed by experiments for sensitivity to the number of cloud-free composites and to different classification legends using crop type classifications. It implied the linear interpolation algorithm cannot reconstruct reliable time series for land cover classifications and historical use of such method is more about mitigating the inability of traditional classifiers to handle missing values rather than improving classifications. Linear interpolation is not necessary for LSTM and Transformer with capability to handle missing values. The training datasets and developed codes in this study are made publicly available.

1. Introduction

Accurate and timely land cover maps are essential to understand terrestrial energy balance, carbon budget, and hydrological cycle (Anderson, 1976; Hansen et al., 2000; Friedl et al., 2002; Zhu and Woodcock, 2014; Wulder et al., 2018; Zhang et al., 2021). In recent decades, the free availability of medium resolution satellite observations provides users with valuable Earth observations to produce land cover products. Satellite time series observations have shown advantages over single-date images in accurate land cover classification (Carrão et al., 2008; Zhu et al., 2012; Bellón et al., 2017; Inglada et al., 2017; Zhang and Roy, 2017; Preidl et al., 2020; Li et al., 2020). This is primarily

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because time series data can capture differences of seasonal spectral changes among land cover types, e.g., phenological characteristics.

Landsat and Sentinel-2 time series observations have been widely used for land cover classification. Landsat series satellites managed by United States Geological Survey (USGS) and National Aeronautics and Space Administration (NASA) provide the longest record of earth observations at 30 m resolution with a revisit cycle of 16 days for a single satellite (Wulder et al., 2012; Wulder et al., 2022). Sentinel-2 mission developed by the European Space Agency (ESA) consists of Sentinel-2A and 2B satellites constellation with a higher revisit cycle of 5 days at 10–20 m resolutions for terrestrial monitoring. Considering their similar spectral band configuration and spatial resolution, combined use of Landsat and Sentinel-2 data could significantly improve the temporal resolution of terrestrial observations (Li and Roy, 2017). Recently, the NASA Harmonized Landsat and Sentinel-2 (HLS) product has been generated with a temporal resolution of < 3 days by combining Landsat-8/9 and Sentinel-2A/B observations (Claverie et al., 2018), and can provide more time series observations to distinguish different land cover types (Woodcock et al., 2020).

Despite the combination of multiple satellite sensors, HLS data still have missing values due to the contamination of clouds and cloud shadows which is common in medium resolution Landsat and Sentinel-2 time series (Egorov et al., 2019; Yan and Roy, 2020; Liu et al., 2024). The conventional machine learning models usually cannot accurately classify the time series with missing values. A commonly used approach to handle the missing values is to implement linear interpolation based on the nearest observations before and after the time period with missing values. For example, Inglada et al (2017) linearly interpolated the Landsat-8 surface reflectance for the year of 2014 to a 16-day temporal period as inputs for a random forest classifier to produce France land cover map. Defourny et al. (2019) linearly interpolated the monthly composite of Landsat-8 and Sentinel-2 time series for near real time crop type classification over Mali, Ukraine and South Africa. Griffiths et al (2019) linearly interpolated the HLS 10-day composite between early October 2015 and the end of 2016 as inputs for a random forest classifier to produce the crop type and land cover map over Germany. De Luca et al (2022) created monthly composites followed by linear interpolation to reconstruct the cloud-contaminated observations of Sentinel-2 bottom-of-atmosphere reflectance between April 2017 and July 2018 as inputs for a random forest classifier to produce land cover mapping in the Mediterranean region. Tran et al. (2022) linearly interpolated the 10 m Sentinel-2 monthly composites for random forest crop type mapping derived from 30 m training labels for two selected HLS tiles across US.

The deep learning methods have also been used for land cover classification using time series observations. The convolutional neural networks (CNN) that were developed for single image classification have been adapted to classify satellite time series for land cover mapping after linear interpolation of the missing values (Pelletier et al., 2019; Zhong et al., 2019; Rußwurm and Körner, 2020; Wang et al., 2020; Debella-Gilo and Gjertsen, 2021; Lobert et al., 2021; Lange et al., 2022; Mohammadi et al., 2023). It should be noted that the non-CNN deep learning architectures widely used in the language processing fields, such as recurrent neural network (RNN) (Sutton, 1988) and Transformer (Vaswani et al., 2017), were designed with capability to handle missing values. The commonly-used RNN model is the long short-term memory (LSTM) which was introduced to solve the problem of gradients vanishing and exploding issue of the plain RNN using gating mechanisms (Hochreiter and Schmidhuber, 1997). The LSTM uses the LSTM unit output features for a time step as the input of the LSTM unit for the next time step to connect successive time steps and extract time series information. If one time step is missing (not observed), the connection will be bypassed to avoid its contribution to classification. Recently, modified RNN architecture have also been developed to better deal with time series data with missing values (Lipton et al., 2016; Baytas et al., 2017; Che et al., 2018). Transformer was developed for natural language processing (NLP) with advantages over RNNs in terms of both feature extraction by memorizing long time series information and training efficiency by allowing parallel computation (Vaswani et al., 2017). Transformer uses a multi-head self-attention mechanism as basic unit to weigh features across all the time steps for temporal feature extraction. Missing values in the time series can be handled by masking them out, typically achieved by setting their attention weights to extremely small values.

The necessity and efficiency of the linear interpolation to handle the missing values for land cover classification is unknown given that the RNN or Transformer models, unlike traditional machine learning, can directly handle input time series missing values. Most research adapting RNN and Transformer models to classify satellite time series still used the linear interpolation method to reconstruct missing values (Jia et al., 2017; Rußwurm and Körner, 2017; Zhu et al., 2017; Ienco et al., 2019; Interdonato et al., 2019; Rußwurm and Körner, 2018; Tang et al., 2022; Xu et al., 2020). A few other studies have directly used RNN or Transformer to handle time series data with missing values for land cover classification without linear interpolation (Metzger et al., 2021; Zhao et al., 2021; Rußwurm et al., 2023). However, none of these studies compared the classifications with and without linear interpolation to justify the necessity or unnecessity and efficiency of the linear interpolation. Note that the unnecessity is possible as linear interpolation may produce inaccurate result in regions of abrupt change with anthropogenic activities or disasters (i.e., crop management, flood, and fire) (Luo et al., 2018). Further, linear interpolation can be time-consuming for the medium and high spatial satellite time series data processing at national or global scales (Garnot et al., 2022).

The objective of this study is to examine whether the linear interpolation of missing values is necessary to classify single pixel satellite time series using RNN and Transformer. In this study, the 30 m HLS time series acquired over Amur River Basin, China with an area of 890,308 km^2 covering 144 HLS tiles (each 110 $\times$ 110 km^2) was classified using the Bi-LSTM and Transformer models with and without linear interpolation. The training samples were derived by visually interpreting eight land cover types using the Landsat images from Google Earth Engine (GEE). The HLS time series without linear interpolation was derived by compositing the observed HLS time series to produce the 16-day composite time series data with possible missing values in some composite periods. The HLS time series with linear interpolation was then derived by linearly interpolating the composites to reconstruct missing values. The model classification comparisons between the time series composites with missing values and interpolated time series composites with no missing values were undertaken visually and quantitatively to evaluate the usefulness of linear interpolation. The main finding of our study is that linearly interpolating missing values is not needed to classify single pixel satellite time series for land cover mapping using recurrent or attention networks.

2. Data

2.1. Study area

Fig. 1 shows the study area (Amur River Basin) colored by the digital elevation model (DEM) from ASTER GDEM Version 3 (https://asterweb.jpl.nasa.gov). The Amur River Basin located in Northeast China, the tenth largest watershed globally, covers the Heilongjiang, Jilin and Inner Mongolia provinces with an area of 890,308 km^2 (Jia et al., 2020). The Greater and Lesser Khingan ranges, and the Changbai Mountains that are both covered by forest, lie in the north and southeast of the study area, respectively. The Amur River Basin covers the Songnen Plain and the Sanjiang Plain that have extensive agricultural activities. The eastern region has a temperate humid monsoon climate, and the western region is dominated by a continental climate (Fang et al., 2017). The average annual precipitation ranges from 250 to 800 mm, which is mainly concentrated in the coastal zone in the east and gradually decreases towards the west. The annual mean air temperature fluctuates from −8 to 6 ℃ (Ning et al., 2022). The main vegetation types are

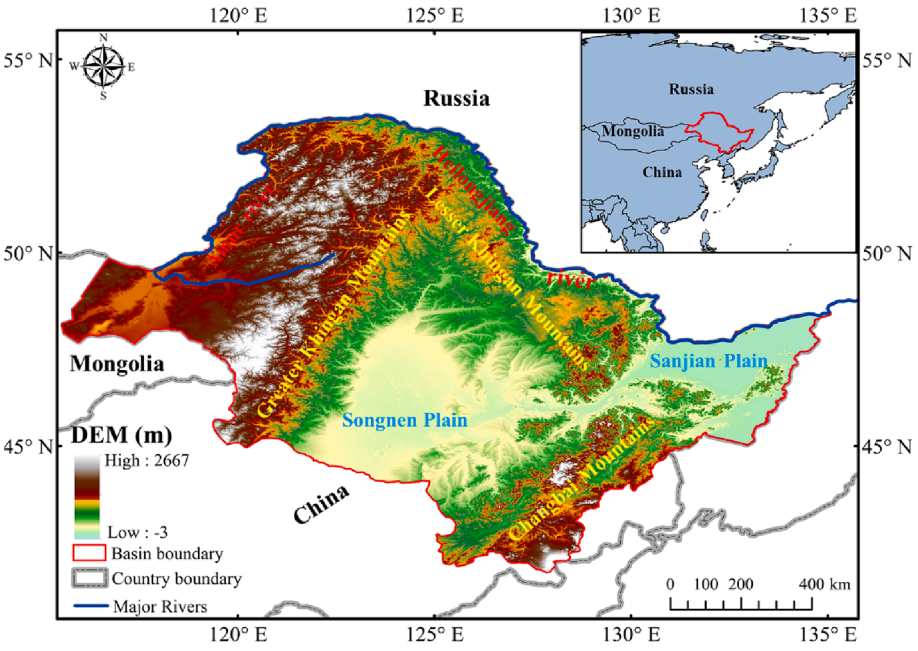

Fig. 1. Location and digital elevation model (DEM) of the study area, Amur River Basin, China.

extensive wetlands, dense forests, vigorous grasslands and agricultural lands (Zhu et al., 2021).

2.2. The NASA Harmonized Landsat Sentinel-2 (HLS) data

We used the NASA Harmonized Landsat Sentinel-2 project (HLS, version 2.0) Landsat 8 Operational Land Imager (OLI, HLSL30) and Sentinel-2 MultiSpectral Instrument (MSI, HLSS30) surface reflectance data. The version 2.0 HLS processing steps include atmospheric correction (Vermote et al., 2016), cloud and cloud-shadow masking (Qiu et al., 2019), spatial co-registration (Gao et al., 2009), bidirectional reflectance distribution function (BRDF) normalization (Roy et al., 2016) and spectral bandpass adjustment (Claverie et al., 2018). HLS images provide moderate spatial resolution (30 m) observations every 1–4 days, depending on latitude (Li and Roy, 2017) by combining Landsat 8/9 OLI sensors (Irons et al., 2012) and Sentinel-2A/B MSI

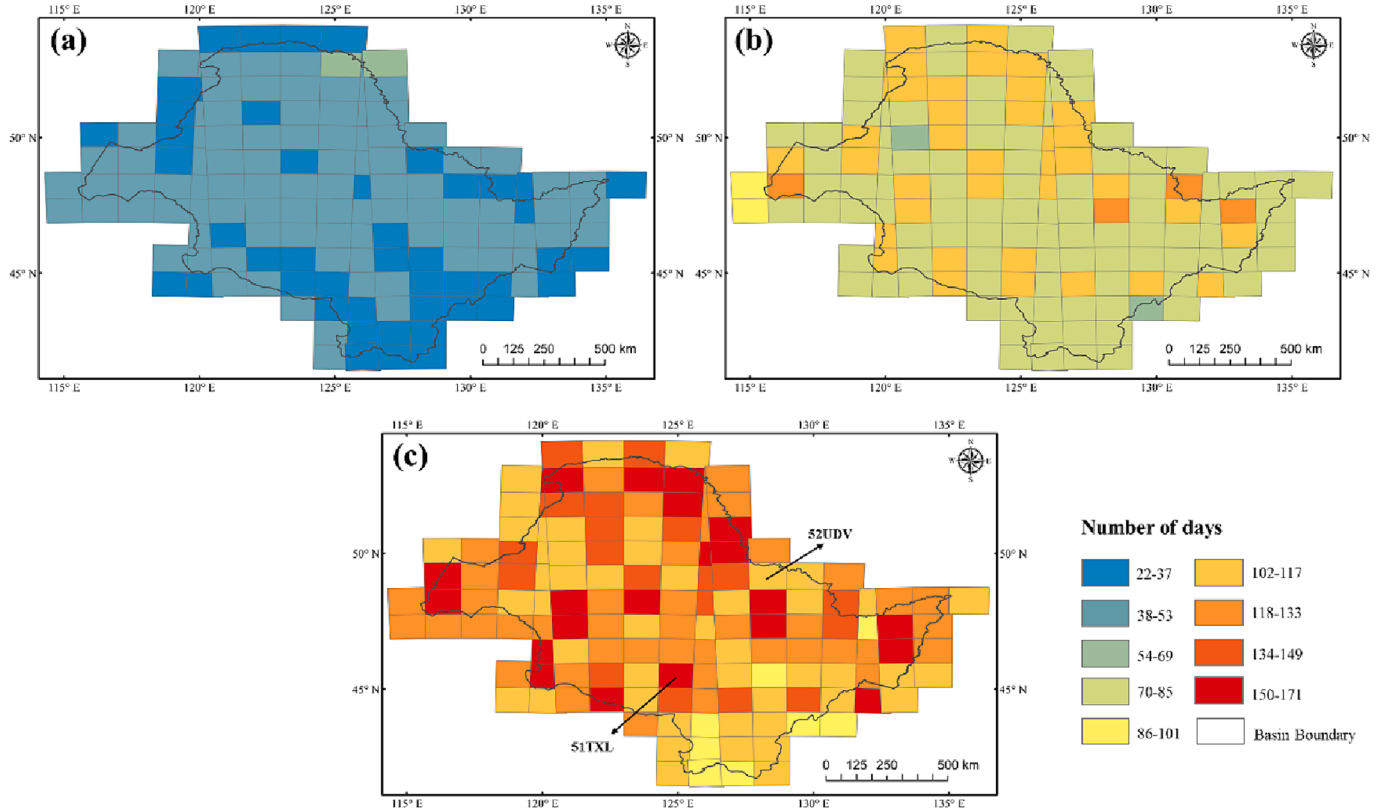

Fig. 2. The number of days in the 2021 growing season that have HLSL30 (a), HLSS30 (b), and the combined HLSL30 and HLSS30 (c) data for each tile.

sensors (Drusch et al., 2012). The HLS uses the Sentienl-2 Universal Transverse Mercator (UTM) Military Grid Reference System (MGRS) tiles with each tile covering 110 × 110 km^2 which usually has horizontal and vertical overlap between two adjacent tiles in the same UTM zone (Roy et al., 2016). For more complete details regarding HLS data, please see Claverie et al. (2018).

In this study, the v2.0 30 m HLS nadir BRDF adjusted reflectance (NBAR) products from the 2021 growing season (6 March 2021 to 15 October 2021) over the 144 HLS tiles with 17,963 images (1940 GB) were used. In 2021 all the two Sentinel-2 sensors and the Landsat 8 sensor are operational with full capacity in data acquisition. Fig. 2 shows the number of days in the 2021 growing season that have HLS data for each tile in the study area. For each tile, there are 22 to 56 days with HLSL30 data, 68 to123 days with HLSS30 data, and a total of 94 to171 combined HLSL30 and HLSS30 images. There are more Sentinel-2 observations than Landsat because Sentinel-2 sensors have higher revisiting cycles compared to Landsat 8 OLI sensor.

2.3. Training and evaluation samples

In this study, the land cover classification system consists of eight types: woodland, grassland, rainfed farmland, paddy field, wetland, water body, built-up land and barren land. Table 1 shows the definition of 30 m pixel samples of each land cover type (Zhu et al., 2021). A stratified random sampling is adopted to generate the 30 m pixel sample locations (Stehman, 1997), which is essential to ensure that sufficient sample sizes are allocated to minority class. First, seven stratifications were produced to signify the possible woodland, grassland, cultivated land, wetland, water body, build-up and barren land using the GlobeLand30 dataset in 2020 (http://www.globallandcover.com/) (Chen et al., 2015). GlobeLand30 is a global land cover product produced by the National Geomatics Centre of China (NGCC) based on the pixel–object–knowledge (POK) classification approach. The dataset has a classification scheme consisting of 10 land cover types including water body, wetland, artificial surface, cultivated land, forest, shrubland, grassland, and barren land, and a spatial resolution of 30 m and global coverage for 2000, 2010, and 2020 with an overall accuracy of 80–85 % (Liu et al., 2022). Subsequently, 1000 samples were randomly selected on each stratum except for the cultivated land stratum where 2000 samples were randomly selected since both the rainfed farmland and paddy field types correspond to the cultivated land stratum. All the selected 8000 samples are taken as the initial sample pool and will be further cleaned using visual interpretation.

Table 1
The number of 30 m pixel samples, the definition and the minimum and mean and maximum Digital Elevation Model (DEM) distribution for each land cover types in the Amur River Basin.

Type	#Pixels	Definition	DEM (m)		
			Min	Mean	Max
woodland	1010	Lands dominated by trees and shrubs	30	640	1466
grassland	1265	Lands with herbaceous types of cover	56	675	1278
rainfed farmland	1087	Agricultural land relying on rainfall for water	35	301	969
paddy field	933	Flooded field of arable land used for growing semiaquatic crops	37	159	542
wetland	1157	Natural wetland covered by vegetation (i.e., swamp, marsh, bog, and fen	33	312	926
water body	737	Permanent water surface	28	254	930
built-up land	860	Land covered by buildings and other impervious structures	37	159	542
barren land	706	Lands has no more than 10 % vegetation coverage	95	243	1380

The selected 8000 samples, i.e., the initial sample pool, were interpreted visually from the satellite images. To interpret the initial sample pool in 2021, Landsat-8 time series images from Google Earth Engine (https://earthengine.google.com/) were used to help visual interpretation. False color composites of Landsat-8 band 5, band 4, and band 3 surface reflectance were used to visually identify the different land cover types. In addition, time series of Normalized Difference Vegetation Index (NDVI) and Normalized Difference Water Index (NDWI) were also plotted to facilitate distinguishing vegetation and water conditions for each sample (see Fig. 3 for example). Six types (woodland, grassland, wetland, water body, built-up land, and barren land) were easily interpreted by visually checking composited images, NDVI and NDWI time series. The rainfed farmland and paddy field are difficult to visually distinguish because the paddy field covered by water is not always observed due to the sparse temporal resolution of Landsat as well as their similar spectral responses. However, NDVI values of paddy field often are slightly higher than that of rainfed farmland because irrigated crops tend to grow better than rainfed ones (Ozdogan and Gutman, 2008), thus paddy field looks darker in the visible bands than rainfed farmland during the growing season. At the same time, there normally exits irrigation water sources near the paddy field such as canals and ponds. Finally, some ambiguous samples were replaced with easy–to-interpret samples.

Table 1 shows the number of samples for each type after the process stated above and the statistics of elevation of the samples derived from DEM. Paddy rice and built-up land are located in low elevations (<160 m) where most human activities were undertaken. In contrast, the woodland and grassland are located in high elevations (>600 m) where climate is more favorable to forest growth and topography makes few disturbances induced by human activities. There is no apparent DEM distribution pattern for other types.

The samples independence should be guaranteed as the spatial correlation of between training and evaluation samples may boost the reported classification accuracy (Colditz. 2015; Kattenborn et al., 2022). Fig. 4 shows the spatial distribution of samples for each type. All the samples are well distributed in the study area despite that grassland and barren land are more concentrated in the west of the study area because the nature of the land cove distribution of the study area (See Fig. 14 in Section 4.4). The between-sample distance is larger than other studies despite that the samples look clustered due to the map generalization issue in Fig. 4. Fig. 5 displays the cumulative histogram of distances of samples to their nearest sample for each land cover type. Among eight land cover types, the minimal sample distance is 0.10 km (3 Landsat pixels) occurring for only a few rainfed farmland pixels. The mean distance ranged from 7.42 km for barren land to 18.6 km for woodland. The average sample distance for all 7735 samples is 10.71 km which is adequate to be considered as spatially independent compared to the existing studies. For example, two CONUS continental scale 30 m Landsat land cover classifications have been derived with the random 2,900,000 and 4,182,823 samples where the average sample distance is 1.99 km and 0.57 km, respectively (Zhang and Roy, 2017, Zhang et al., 2023).

3. Methods

3.1. Composites without (with missing values) and with linear interpolation (without missing values)

The 2021 growing season HLS time series is temporally composited into fixed 16-day periods for land cover mapping (Griffiths et al., 2019; Potapov et al., 2019) and there are 14 16-day periods from 6 March 2021 to 15 October 2021. Temporal compositing was undertaken for each 30 m HLS time series by selecting the best quality observation within each 16-day period. A median NDVI compositing method (Fig. 6) is used which selects the observation that has the median of all the NDVI values from all cloud free observations filtered by the QA layer (Yang

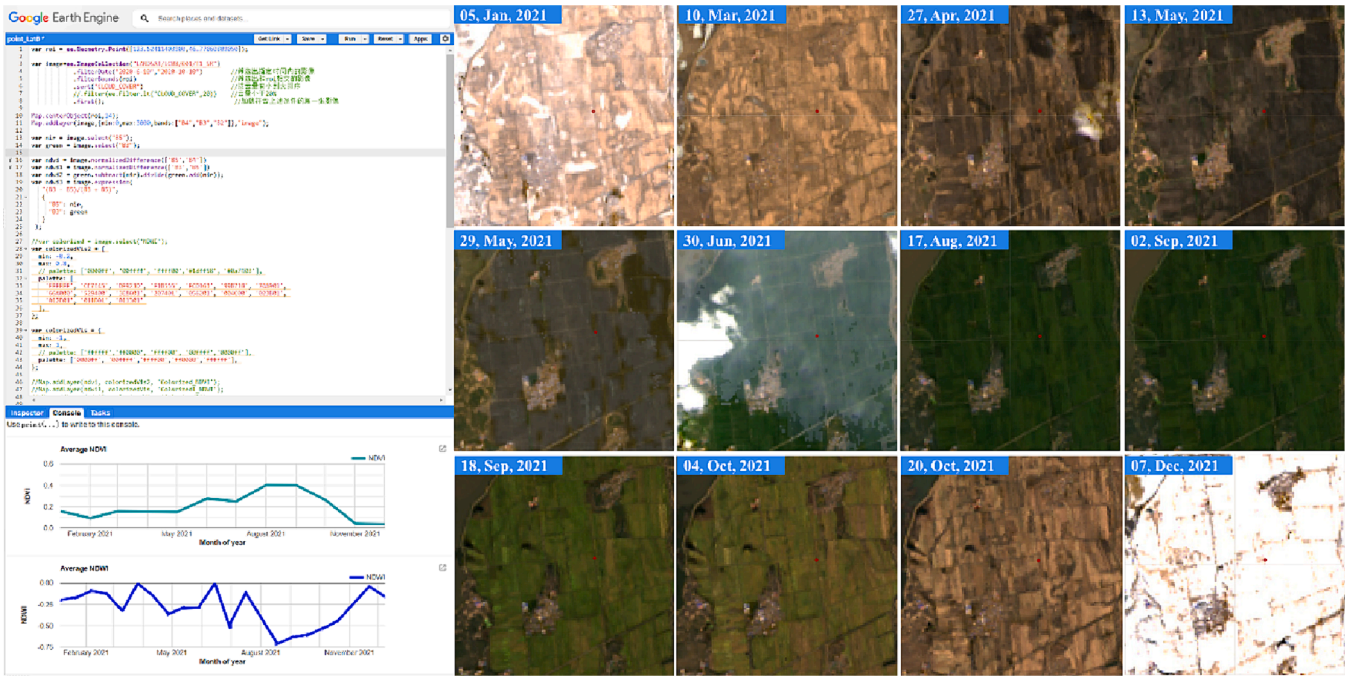

Fig. 3. An example of the display panel of interpretation tool for Landsat-8 images time series from Google Earth Engine at a paddy field location.

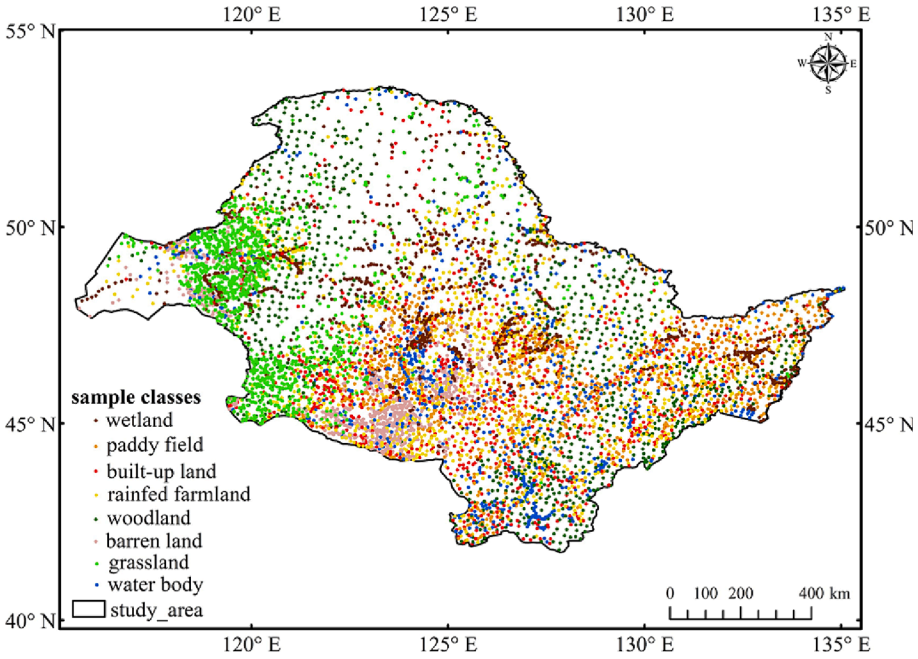

Fig. 4. Samples spatial distribution for each land cover type in the Amur River Basin where the samples are not clustered given the scale of the map (i.e., 890,308 km^2).

et al., 1998; Robinson et al., 2017; Asam et al., 2023). Fourteen predictor variables were used at each compositing period including seven band surface reflectance and seven spectral band indices. Specifically, the seven surface reflectance data included the coastal/aerosol, blue, green, red, near infrared (NIR), and two shortwave infrared (SWIR) bands. The coastal/aerosol spectral band was used since it was proved to be useful for inland water mapping and water quality studies despite of subject to more atmospheric contaminations than the other spectral bands (Olmanson et al., 2016; Aroma et al., 2024). The seven spectral band indices related with vegetation greenness and water content included the Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), Soil Adjusted Vegetation Index (SAVI), Modified Soil Adjusted Vegetation Index (MSAVI), Normalized Difference Moisture Index (NDMI), Normalized Difference Water Index (NDWI) and Normalized Difference Infrared Index-band 7 (NDIIb7) calculated from band 1–7 surface reflectance. The indices were calculated using the equation in Table 2.

The composites derived above can still have missing values (set as −9999 in this study) in certain 16-day periods when there are no good quality observations during a 16-day period. For example, there is no good quality observations in the entire 16-day period due to cloud cover despite of < 3 day revisit cycle of the HLS data. In addition, we generate

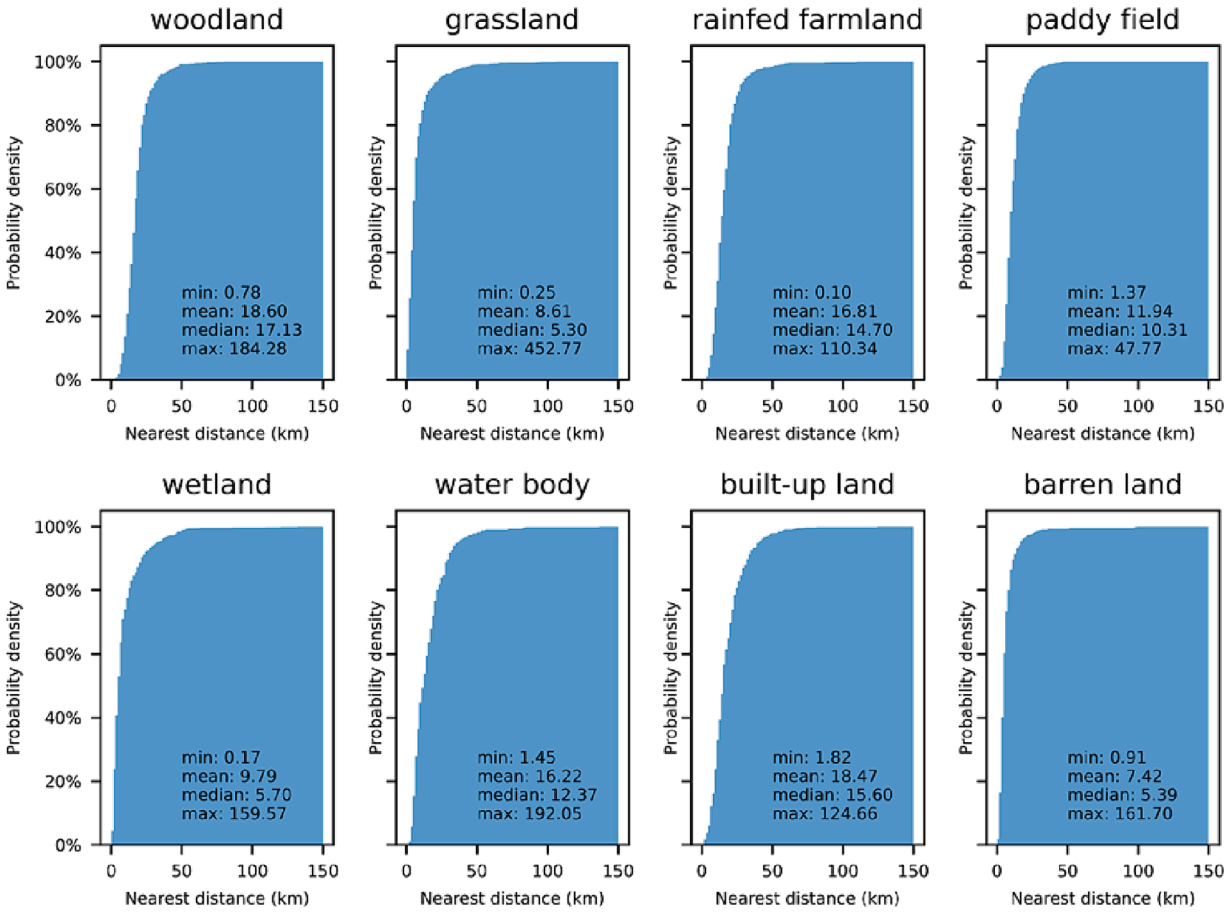

Fig. 5. Cumulative histogram of distances of samples to their nearest samples for eight land cover types.

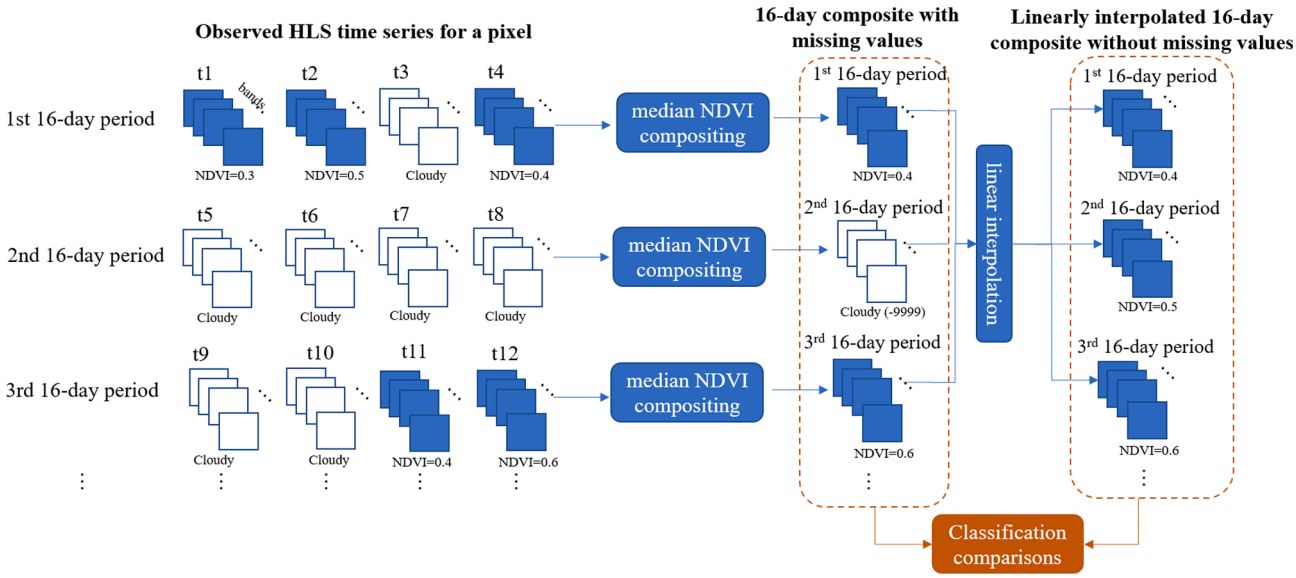

Fig. 6. A pixel example of 16-day composites without (with missing values) and with linear interpolation (without missing values) for classification comparisons. t_1, t_2, t_3, …., t_{12} represent the HLS data acquisition dates in first three out of the 14 16-day compositing periods (i.e., 1st, 2nd, and 3rd).

another set of composites with no missing values by linear interpolation of missing values (Inglada et al., 2017; Tran et al., 2022). If two valid reflectance composites (ρ_i^b and ρ_i^a) exist before and after the missing value composite 16-day period t for band i, ρ_i^b and ρ_i^a, and the corresponding 16-day period periods (t_b and t_a) are used to calculate an interpolated reflectance value for the missing value composite t.

$$\rho_i^t = \rho_i^b + \frac{\rho_i^a - \rho_i^b}{t_a - t_b}(t - t_b) \quad (1)$$

where t_b, t_b and t are ≥ 1 and ≤ 14 (i.e., the total number of the 16-day

Table 2
Descriptions of the fourteen HLS predictor variables. Band8A NIR band is used as it is more similar than the band8 to the Landsat NIR band.

Type	Variables	Description	Reference
Surface reflectance	ρ_{cosstal}	Coastal/aerosol band surface reflectance	(Claverie et al., 2018)
	ρ_{blue}	Blue band surface reflectance	
	ρ_{green}	Green band surface reflectance	
	ρ_{red}	Red band surface reflectance	
	ρ_{nir}	NIR band surface reflectance	
	ρ_{swir1}	SWIR1 band surface reflectance	
	ρ_{swir2}	SWIR2 band surface reflectance	
Vegetation indices	NDVI	$\frac{\rho_{\text{nir}}-\rho_{\text{red}}}{\rho_{\text{nir}}+\rho_{\text{red}}}$	(Tucker, 1979)
	EVI	$\frac{\rho_{\text{nir}}-\rho_{\text{red}}}{\rho_{\text{nir}}+6\times\rho_{\text{nir}}-7.5\times\rho_{\text{red}}+1}$	(Huete et al., 2002)
	SAVI	$1.5\times\frac{\rho_{\text{nir}}-\rho_{\text{red}}}{\rho_{\text{nir}}+\rho_{\text{red}}+0.5}$	(Huete, 1988)
	MSAVI	$\frac{2\times\rho_{\text{nir}}+1-\sqrt{(2\times\rho_{\text{nir}}+1)^2-8\times(\rho_{\text{nir}}-\rho_{\text{red}})}}{2}$	(Qi et al., 1994)
Waterindices	NDMI	$\frac{\rho_{\text{nir}}-\rho_{\text{swir1}}}{\rho_{\text{nir}}+\rho_{\text{swir1}}}$	(Wilson and Sader, 2002)
	NDWI	$\frac{\rho_{\text{green}}-\rho_{\text{nir}}}{\rho_{\text{green}}+\rho_{\text{nir}}}$	(McFeeters, 1996)
	NDIIb7	$\frac{\rho_{\text{nir}}-\rho_{\text{swir2}}}{\rho_{\text{nir}}+\rho_{\text{swir2}}}$	(Hunt and Rock, 1989)

temporal periods). The Eq. (1) is applicable if both ρ_i^b or ρ_i^a are valid. If only one valid reflectance composite (ρ_i^b or ρ_i^a) exists around the target 16-day period t for band i, the missing value composite ρ_i^t will be set as ρ_i^b or ρ_i^a.

The "*composites with missing values*" and "*linearly interpolated composites*" were referred to as composites without and with linear interpolation, respectively. We should note, however, the "*linearly interpolated composites*" are different from the completely observed time series composites (not existing) despite both have no gaps in time series.

3.2. Classification models

Linear interpolation of missing values is often needed for land cover classification in traditional machine learning algorithms (Kim et al., 2014; Cai et al., 2018; Defourny et al., 2019; Griffiths et al., 2019; Pelletier et al., 2019; Liu et al., 2020; De Luca et al., 2022). This is mainly because traditional machine learning classifiers cannot mask out missing values in classification and directly classifying data with missing values will degrade the classification model performance (Breiman, 2001; Stekhoven and Bühlmann, 2012). With the advance of deep learning models that can handle missing data (i.e., mask out missing values in the model), the contribution of linear interpolation should be examined. This study is to evaluate two approaches to handle the missing value issue. (i) The first approach is a traditional linear interpolation approach to reconstruct the missing values before fed into deep learning classification models. (ii) The second approach directly took advantage of the capability of handling missing values of the deep learning classification models by masking out missing values. The time series composite with missing values can be directly used as input for deep learning classification models.

Two types of deep learning models originated from the machine translation that can mask out missing values in classification were used in this study, which are based on the bi-directional Long Short Term Memory (Bi-LSTM) model (Graves and Schmidhuber, 2005) and the Transformer model (Vaswani et al., 2017). The CNNs that has been used for time series land cover classification (Pelletier et al., 2019; Zhang et al., 2023) was not used as CNN usually assigns missing values as zero rather than masking out them in classification (known as zero-padding) (Zhao et al., 2021). Consequently, four land cover classification models were used (i) Bi-LSTM to classify time series composites without linear interpolation (i.e., with missing values); ii) Bi-LSTM to classify time series composites with linear interpolation (i.e., with no missing values); (iii) Transformer to classify time series composites without linear interpolation; and iv) Transformer to classify time series composites with linear interpolation.

In this study, we inserted a masking layer to the bi-directional LSTM (Bi-LSTM) architecture (Graves and Schmidhuber, 2005) to inform the model which time periods have missing values in a time series. Bi-LSTM includes a forward (LSTM from t to $t+1$) and backward (LSTM from $t+1$ to t) flow simultaneously that are usually considered to produce higher classification accuracy. The output features from the forward and backward flows can be combined to derive the final hidden feature output in several ways such as average, sum, multiplication, or concatenation. In this study, concatenation is used, and we found little difference among different ways. Bi-LSTM model takes a pixel time series of 16-day compositing values during the growing season as input (14×14), and yields a predicted type label $c \in \{0, 1, \cdots l-1\}$ given the number of types l ($l = 8$ in this study) (Fig. 7). Each training sample $X \in \mathbb{R}^{T\times m}$ is a multivariate time series with length T ($T = 14$ in this study) and with each temporal period having m variables ($m = 14$ in this study), i.e., $X \in \mathbb{R}^{T\times m} = [x_1, x_2, \cdots, x_T]$. $x_t \in \mathbb{R}^m$ represents the t-th 16-day compositing values of all variables and x_t^d denotes d-th variable of x_t. To handle the missing values from 16-days compositing time series, a masking vector $m_t \in \{0,1\}^m$ in the masking layer is introduced to indicate which 16-day temporal periods are missing:

$$m_t = \begin{cases} 0, if\, x_t is missing \\ 1, otherwise \end{cases} \quad (2)$$

when $m_t = 0$ is detected in the LSTM cell, the input x_t will be skipped for processing. Specifically, cell state and hidden state features at the time step t-1 will be directly transferred to the LSTM cell at the time step $t+1$. Four hidden Bi-LSTM layers are cascaded among which dropout regularization layers are added (Dahl et al., 2013). A fully connected layer with a rectified linear unit (ReLU) activation function (Glorot et al., 2011) followed by a batch-normalization layer (Ioffe and Szegedy, 2015) are used after the last Bi-LSTM layer. The output layer is a fully connected layer with as many neurons as the number of land cover types by applying a Softmax activation function (Zhang et al., 2016) to produce the classification probability for each land cover type.

In terms of Transformer model, we directly adopted the encoder module of the classic Transformer architecture proposed by Vaswani et al., (2017), but modified its positional encoding function, inserted a masking layer to mask out missing values of time series and added a masked averaged layer before the classification. We used positional encodings with learnable model coefficients rather than using sine and cosine functions with pre-defined parameters (i.e., without learnable

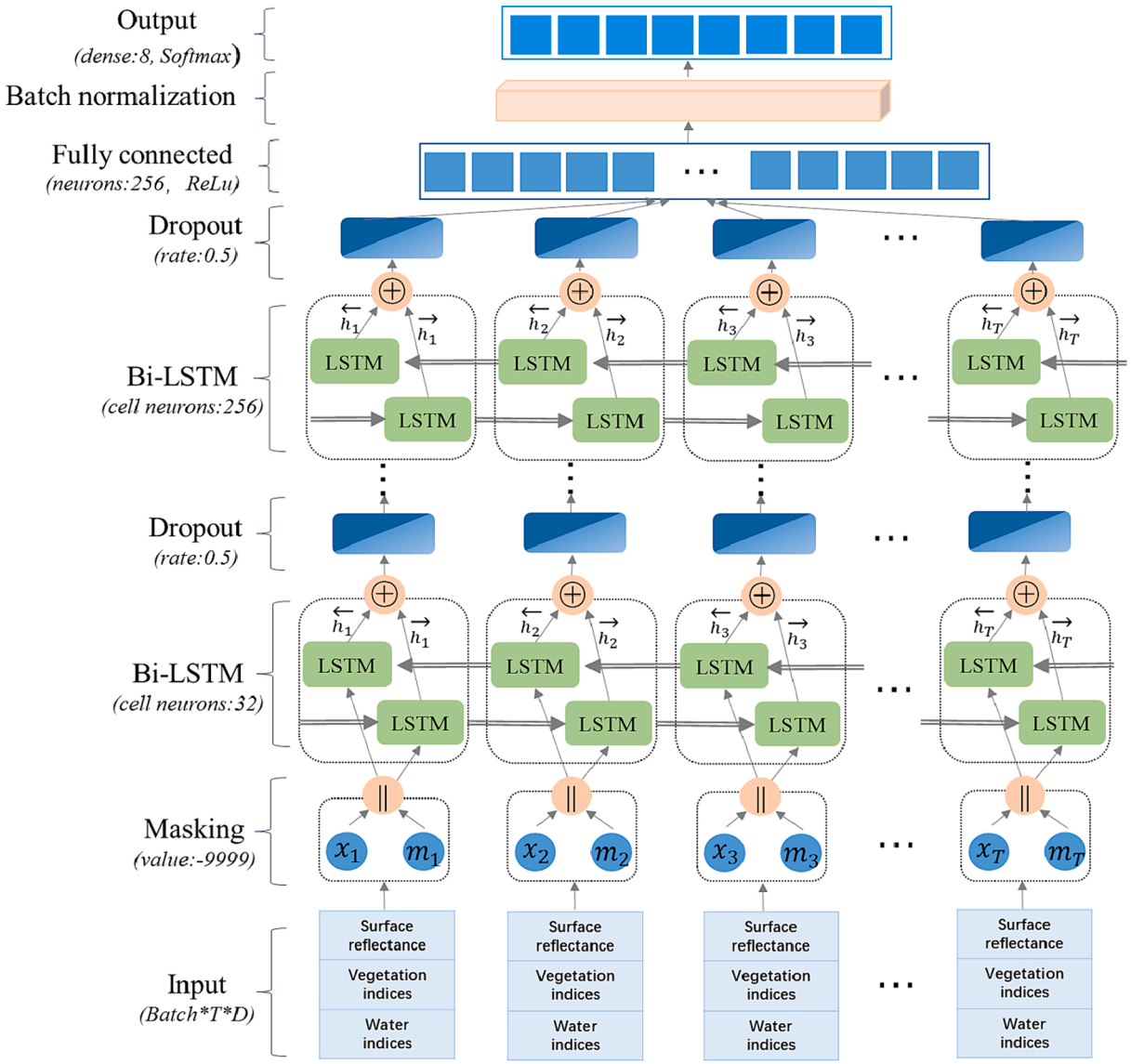

Fig. 7. Architecture of Bi-LSTM land cover classification. The masking layer is optional and only used to classify the time series composites with missing values.

coefficients) in Vaswani et al., (2017) as we found the former produced better classification accuracy. Specifically, both reflectance x_t and its time series position $x_t^p = \frac{t}{T}$ are linearly projected using learnable matrix and bias to derive two feature vectors with the same dimension d which are them summed to derive the projected feature vector u_t. The time series feature vector u_t ($t = 1, 2, \ldots, T$) is fed into the encoder of the Transformer architecture as described in Vaswani et al., (2017). The core of Transformer is the attention layer to derive the transformed feature vector for each temporal period by weighting sum of all the input time series feature vectors. In the classification of composites with missing values, we masked the composite time series $x_t \in \mathbb{R}^m (t = 1, 2, \cdots, 14)$ using a masking vector $m_t \in \{0, 1\}^m$ and set the attention weights over masked operations to extremely low values to ignore their contribution. Before the derivation of classification probability, we used a masked average layer to derive an average feature vector from the time series feature vectors at the unmasked 16-day temporal periods as did in Zhang et al. (2024) (Fig. 8).

3.3. Model training and evaluation

All the Bi-LSTM and Transformer models were trained using the cross-entropy loss function. To avoid convergence issues during training, the fourteen input variables x_t were standardized using dimension-specific mean and standard deviation derived from the training samples. We trained the Bi-LSTM and Transformer models using the Adam optimizer (Kingma and Ba, 2014). The Bi-LSTM optimal hyper-parameters were determined by sampling candidate hyper-parameters of the batch size $b \in \{16, 32, 64, 128\}$, learning rate $\alpha \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, dropout $\rho_{drop} \in \{0, 0.5, 0.8\}$, epochs $\in \{40, 80, 120\}$, dimension d of hidden features $\in \{32, 64, 128, 256\}$, and neurons for a fully connected layer $\in \{20, 40, 60, 100\}$. Thus, each model was run for $4 \times 5 \times 3 \times 3 \times 4 \times 4$ times to find the optimal combination. The optimal b, α, ρ_{drop}, epochs and neurons are 32, 10^{-4}, 0.5, 80 and 100, respectively. The dimension d of hidden features was set as 32, 64, 126, 256 for four layers, respectively. The similar strategy was applied to Transformer model, and the optimal b, α, ρ_{drop}, epochs and number of heads $\in \{1, 2, 3, 4, 5, 6\}$ was 32, 10^{-4}, 0.1, 80 and 4, respectively. Overall, the training time of finding the optimal parameters configures is roughly 15 days and 2 days for the Bi-LSTM and Transformer models, respectively.

The classification model was evaluated using the k-fold cross-validation ($k = 5$). Specifically, the 7755 samples were partitioned

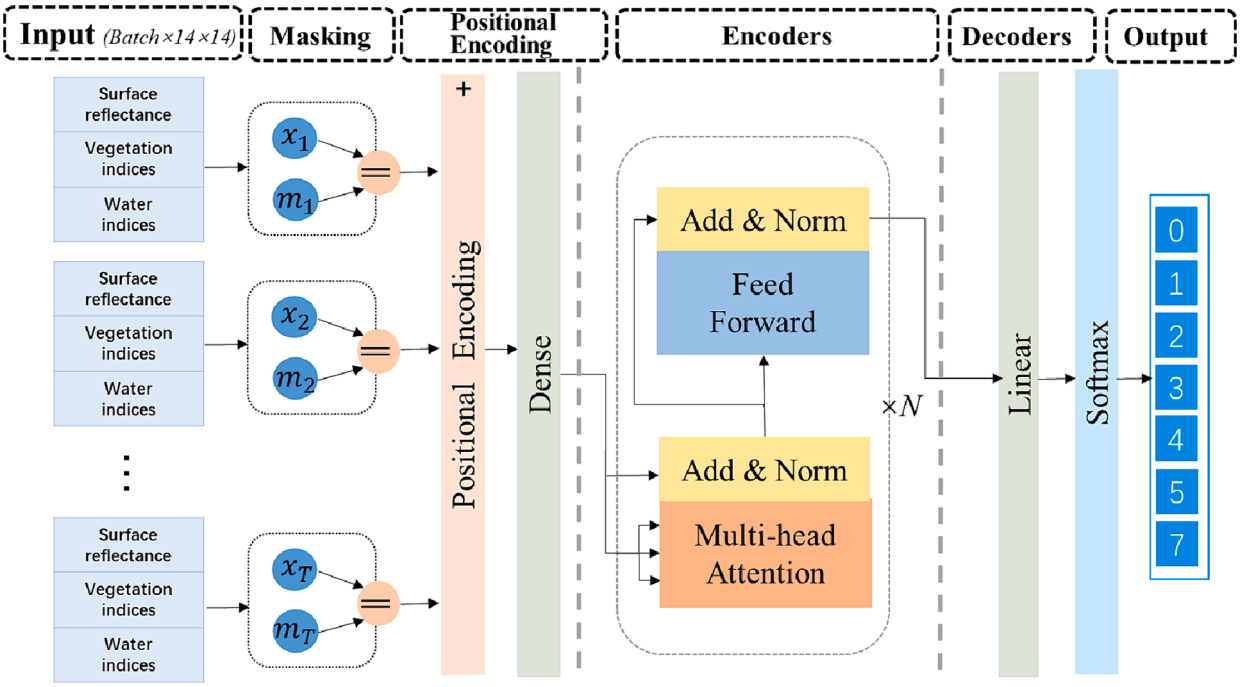

Fig. 8. Architecture of Transformer based land cover classification. The masking layer is optional and only used to classify the time series composite with missing values.

into five subsets with the same number of samples (7755/5 = 1551). One subset is used for the testing dataset, and other four subsets was split into the training dataset for model parameters learning and validation dataset for parameters optimization (training and validation sample ratio of 24:1). This was undertaken five times so that each single subset was used as testing dataset. The overall accuracy (OA) and F1-score of the testing dataset were derived. The F1 score is defined as the harmonic mean of the user's accuracy and producer's accuracy. Each model was run for five times to examine the deep learning model randomness for each fold training and testing. This, combined with five-fold cross-validation, mean each model was trained 25 times using the optimal hyper-parameters.

Land cover maps for the entire region and for two example HLS tiles (52UDV and 51TXL) using the four classification models were visually and quantitatively compared for assessment. The two tiles were selected as they represent the relatively infrequent and frequent of HLS data, respectively, as shown in Fig. 2(c).

The computation time to classify the two examples tile 52UDV and 51TXL tile composite time series without and with linear interpolation were compared. Note computation time was not counted for all the 144 tiles in the study area due to the tremendous additional computation time for composite with linear interpolation (See results and discussion). The process was performed on a Linux workstation (Ubuntu 16.04 LTS) with two Intel Xeon Gold 6230 Processors (2.1G/ 20 Core/27.5 M), 128 GB of RAM, and two NVIDIA GeForce RTX 3060 Ti graphics cards (each 11 GB memory). All the models were implemented on the Python platform using the Keras library.

4. Results

4.1. Time series characteristics of NDVI for different land cover types

Table 3 shows annual percentages of the 16-day composites with missing values for each type samples and Fig. 9 shows the missing value percentages for each 16-day composite period. There were on average 15.35 % 16-day composites with missing values. Water body type had the most missing values perhaps because water body can be misidentified as cloud shadow in HLS. Fig. 8 shows the first 16-day composite period (from March 6 to March 21) had the most missing values because of snow and ice cover in northern latitude and in mountain area. Furthermore, the 16-day composites periods from number 8 to 11 (from June 26 to August 28) in summer had more missing values due to the more frequent cloud cover of the rainy season.

Fig. 10 shows time series mean and standard deviations of 16-day time series of NDVI for each land cover derived from all the time series samples without (Fig. 10 top) and with (Fig. 10 middle) linear interpolation as well as the relative percentage differences between mean NDVI composites of samples without and with linear interpolation for each land cover (Fig. 10 bottom). The two sets of time series mean and standard deviations derived from composites with and without linear interpolation exhibit very similar patterns. However, it is hard to clarify whether the time series difference before and after linear interpolation for single pixel is significant or not since the real observations at the interpolation time steps were unknown (See Section 5.1). NDVI time series from water body have the lowest values and largest standard deviations, and they are far away from other land cover types. By contrast, woodland including forest and shrubs have the highest NDVI values and the longest greenness period. NDVI values from built-up land are slightly higher than barren land due to the vegetation plantation

Table 3
The annual percentage of 16-day composites with missing values for samples in each type.

Type	woodland	grassland	paddy field	rainfed farmland	wetland	water body	built-up land	barren land
Percentage(%)	20.09	9.27	13.63	12.08	12.50	29.60	14.64	17.43

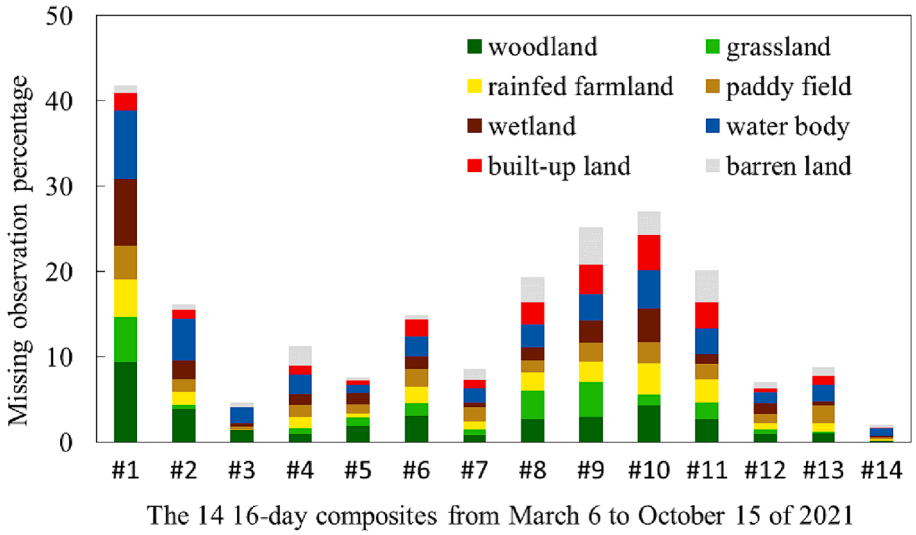

Fig. 9. The missing value pattern for 14 16-day composite periods from March to October of 2021 (#1 indicates March 6 to March 21, #2: March 22 to April 6, #3: April 7 to April 22, #4: April 23 to May 8, #5: May 9 to May 24, #6: May 25 to June 9, #7: June 10 to June 25, #8: June 26 to July 11, #9: July 12 to July 27, #10: July 28 to August 12, #11: August 13 to August 28, #12: August 29 to September 13, #13: September 14 to September 29, #14: September 30 to October 15).

around roads or built-up regions. The similar patterns were also found between rainfed farmland and paddy field and between wetland and grassland. However, paddy field has higher NDVI values during the growing season compared to rainfed farmland, and wetland appears greener than grassland during the early growing season, possibly due to more water availability in wetland. All relative percentage differences between mean NDVI composites of samples without and with linear interpolation from DOY 88 to 280 for all land covers are lower than 10 % except for water bodies.

4.2. Accuracy comparison of classifications without and with linear interpolation

Tables 4 and 5 show Bi-LSTM and Transformer models overall accuracy from the testing datasets for classification of time series composite without and with linear interpolation, respectively. The mean overall accuracy differences between classifications of composites without and with linear interpolation was < 0.2 % for Bi-LSTM models and < 0.5 % for Transformer models. The mean overall accuracy differences were lower than corresponding standard deviation of the overall accuracies by running model five times. This demonstrates that linear interpolation contributes little improvement for land cover classification using the Bi-LSTM and Transformer models. The mean overall accuracies for Bi-LSTM classification of time series composite without linear interpolation ranged from 90.0 % to 90.4 % while those with linear interpolation ranged from 90.2 % to 90.4 %. The mean overall accuracies were 89.8 % to 90.3 % for Transformer classification of composite without linear interpolation, and 89.7 % to 90.2 % for those with linear interpolation. In addition, the median overall accuracies from the 25 Bi-LSTM models were slightly higher than those of Transformer models. The standard derivation of the overall accuracies within a specific fold ranging from $\pm$ 0.1 % to $\pm$ 0.4 % (the last row) were smaller than that across the five folds ranging from $\pm$ 0.6 % to $\pm$ 1.0 % (the last column). This was expected as the standard derivation within a specific fold reflects only randomness in deep learning model training and the standard derivation values across the five folds reflects randomness in both sample split and deep learning model training.

4.3. Land cover specific accuracy comparison of classifications without and with linear interpolation

Table 6 shows the mean and standard deviation of the testing dataset F1 values for each land cover using Bi-LSTM and Transformer models run for five times. Note that for each run of the five-fold cross validation, a single F1 score is derived by considering all the five-fold testing dataset, i.e., all the samples. In this way, the standard deviations of the F1 scores of the five time runs only reflect the randomness in the model training (not in the training and testing data split) so that they can be used as a control to indicate whether the accuracy differences among classification models are pronounced. The mean F1 values of eight land cover types have little difference for classification of composites without and with linear interpolation with < 0.007 for Bi-LSTM models and < 0.008 for Transformer models. Furthermore, all mean F1 difference values between classifications of composites without and with linear interpolation for wetland, woodland, grassland, water body, rainfed farmland, build-up and barren land classes are lower than the corresponding standard derivations. The results indicate there is no significant accuracy difference for classification of composites without and with linear interpolation.

Additionally, the built-up land type has the highest classification accuracy with the highest F1 values and wetland has the lowest F1 values for both Bi-LSTM and Transformer models for all the land cover types. The paddy field type ranks second in F1 values for Bi-LSTM model classification, while the water body type has second highest F1 values for Transformer model classification. The classification accuracy of grassland and rainfed farmland types for Bi-LSTM and Transformer models have similar F1 values < 0.9.

4.4. Classification map comparison

Four trained models with the median overall accuracy values (bold values in Tables 4 and 5) were used for land cover mapping. Classification maps from two HLS tiles (52UDV and 51TXL) were visually examined to compare differences between classifications of composites without and with linear interpolation. Figs. 11 and 12 show classification results for tile 52UDV and 51TXL, respectively, which have the relatively infrequent and frequent HLS data. Figs. 13 and 14 show the corresponding zoom-in details for areas with the 500 $\times$ 500 30 m pixels. In general, the classification maps were visually comparable/similar between classifications of composites without and with linear interpolation for both Bi-LSTM and Transformer models.

91.80 % of tile 52UDV pixels are classified as the same class between classifications of composites without and with linear interpolation by the Bi-LSTM model and 89.05 % by the Transformer model. And 91.38

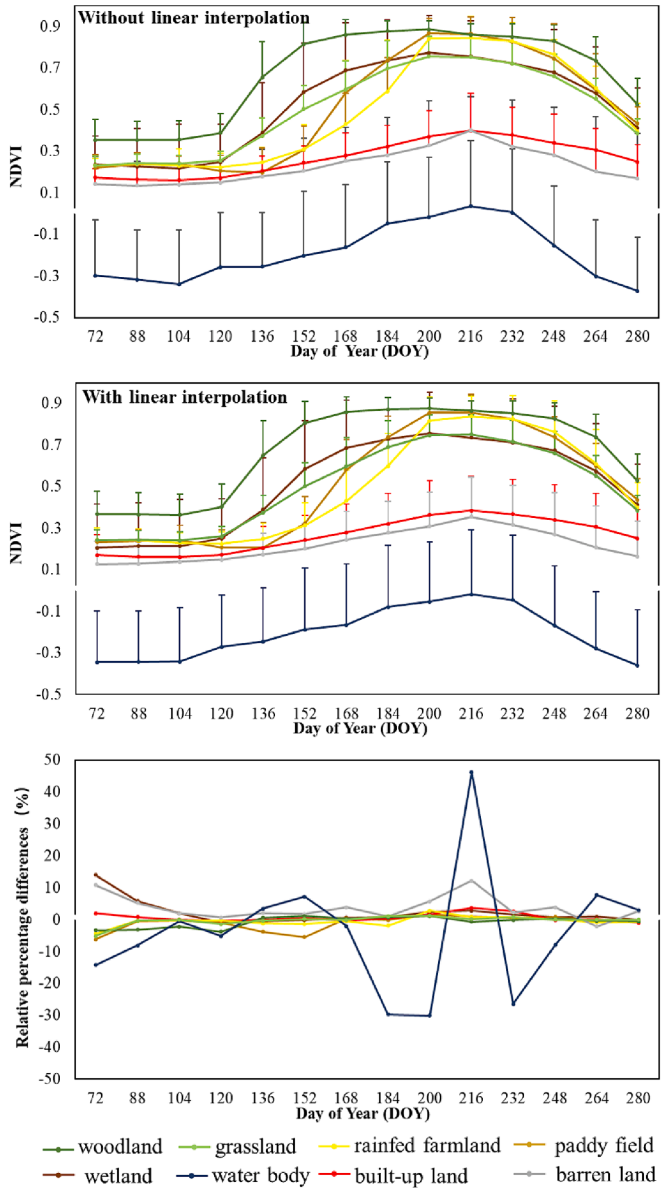

Fig. 10. Time series mean (curves) and standard deviations (vertical lines) of NDVI for samples, and the relative percentage differences (%) between mean NDVI composites of samples without and with linear interpolation for each land cover (Table 1).

% and 87.40 % (for Bi-LSTM and Transformer model) of the tile 51TXL pixel are classified as the same between classifications of composites without and with linear interpolation. These indicated a relatively high classification consistency between without and with linear interpolation.

Fig. 15 shows the classification maps for the entire study area with 144 HLS tiles from 17,963 images with a dataset size of nearly 1940 GB derived using Bi-LSTM (left) and Transformer (right) models for image composites without linear interpolation. The classification maps using image composites with linear interpolation were not derived given the additional computation time of linear interpolation (See Section 4.5) and little difference were found between them (See results in Section 4.1-4.3). Fig. S1 shows the proportion distribution of each land cover of the Bi-LSTM and Transformer model classification maps. Both the land cover maps (Fig. 15) and proportions (Fig. S1) are similar between the Bi-LSTM and Transformer models. Specifically, the study area has woodland with the largest proportion of 40.95 % and 37.84 % for Bi-

Table 4
Bi-LSTM testing dataset overall accuracy (%) derived for composites without and with linear interpolation using 5-fold cross-validation. The mean and standard deviation (SD) in the last column were derived from all the 5-fold overall accuracies for each run and the mean and SD in the last row were derived from all the five runs for each fold. The two bold values were the median from 25 (5-fold cross-validation × five times) models for the composite with and without linear interpolation, respectively.

Run	Linear interpolation	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean ± SD
#1	without	90.1	91.2	89.9	91.4	88.9	90.3 ± 1.0
	with	89.9	91.7	89.4	90.7	89.3	90.2 ± 1.0
#2	without	90.7	91.2	90.0	90.8	89.3	90.4 ± 0.6
	with	**89.9**	91.9	89.5	91.0	89.7	90.4 ± 1.0
#3	without	**90.3**	91.4	89.3	90.5	88.7	90.0 ± 1.1
	with	89.7	91.6	89.4	90.7	89.6	90.2 ± 0.9
#4	without	89.6	91.7	89.7	91.1	89.2	90.3 ± 1.0
	with	90.1	91.7	89.7	90.6	89.7	90.4 ± 0.7
#5	without	90.3	91.2	89.7	90.7	89.4	90.3 ± 0.7
	with	90.2	91.7	89.6	90.8	89.6	90.4 ± 0.9
Mean ± SD	without	90.2 ± 0.4	91.3 ± 0.2	89.7 ± 0.3	90.9 ± 0.4	89.1 ± 0.3	
	with	90.0 ± 0.2	91.7 ± 0.1	89.5 ± 0.1	90.8 ± 02	89.6 ± 0.2	

Table 5
Same as Table 4 but for Transformer models.

Run	Linear interpolation	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean ± SD
#1	without	88.8	91.3	89.0	91.5	88.6	89.8 ± 1.4
	with	88.9	90.7	88.5	90.9	89.6	89.7 ± 1.1
#2	without	88.9	91.7	90.4	91.4	88.7	90.2 ± 1.4
	with	89.0	90.5	89.3	91.3	89.0	89.8 ± 1.0
#3	without	89.1	91.2	89.0	91.0	88.8	89.8 ± 1.2
	with	89.3	91.8	89.5	91.6	89.0	90.2 ± 1.3
#4	without	88.8	90.8	89.0	91.4	88.8	89.8 ± 1.2
	with	89.0	91.3	89.6	91.2	88.8	90.0 ± 1.2
#5	without	90.4	91.3	**89.3**	91.3	89.0	90.3 ± 1.1
	with	**89.5**	91.0	89.2	90.3	89.1	89.8 ± 0.8
Mean ± SD	without	89.2 ± 0.7	91.3 ± 0.3	89.3 ± 0.6	91.3 ± 0.2	88.8 ± 0.2	
	with	89.1 ± 0.3	91.1 ± 0.5	89.2 ± 0.4	91.1 ± 0.5	89.1 ± 0.3	

LSTM and Transformer models which is followed by rainfed farmland with 22.88 % (Bi-LSTM) and 21.95 % (Transformer). Obviously, the water body accounts for the smallest percentages of 1.87 % (Bi-LSTM) and 1.75 % (Transformer). The largest difference between Bi-LSTM and Transformer models is for wetland where the corresponding percentage are 8.93 % (Bi-LSTM) and 11.82 % (Transformer), respectively.

Table 6
F1 values of the testing sets for each land cover for Bi-LSTM and Transformer models. The mean and standard deviation of five times run are shown.

Type	Bi-LSTM		Transformer	
	without linear interpolation	with linear interpolation	without linear interpolation	with linear interpolation
woodland	0.915 ± 0.004	0.917 ± 0.001	0.920 ± 0.007	0.921 ± 0.005
grassland	0.886 ± 0.003	0.885 ± 0.002	0.880 ± 0.005	0.883 ± 0.006
paddy field	0.942 ± 0.001	0.939 ± 0.001	0.927 ± 0.003	0.932 ± 0.002
rainfed farmland	0.887 ± 0.004	0.894 ± 0.004	0.876 ± 0.005	0.884 ± 0.006
wetland	0.827 ± 0.003	0.825 ± 0.002	0.831 ± 0.008	0.836 ± 0.004
water body	0.937 ± 0.002	0.940 ± 0.003	0.940 ± 0.005	0.941 ± 0.003
built-up land	0.956 ± 0.003	0.960 ± 0.003	0.962 ± 0.002	0.963 ± 0.002
barren land	0.906 ± 0.004	0.905 ± 0.006	0.908 ± 0.001	0.908 ± 0.003

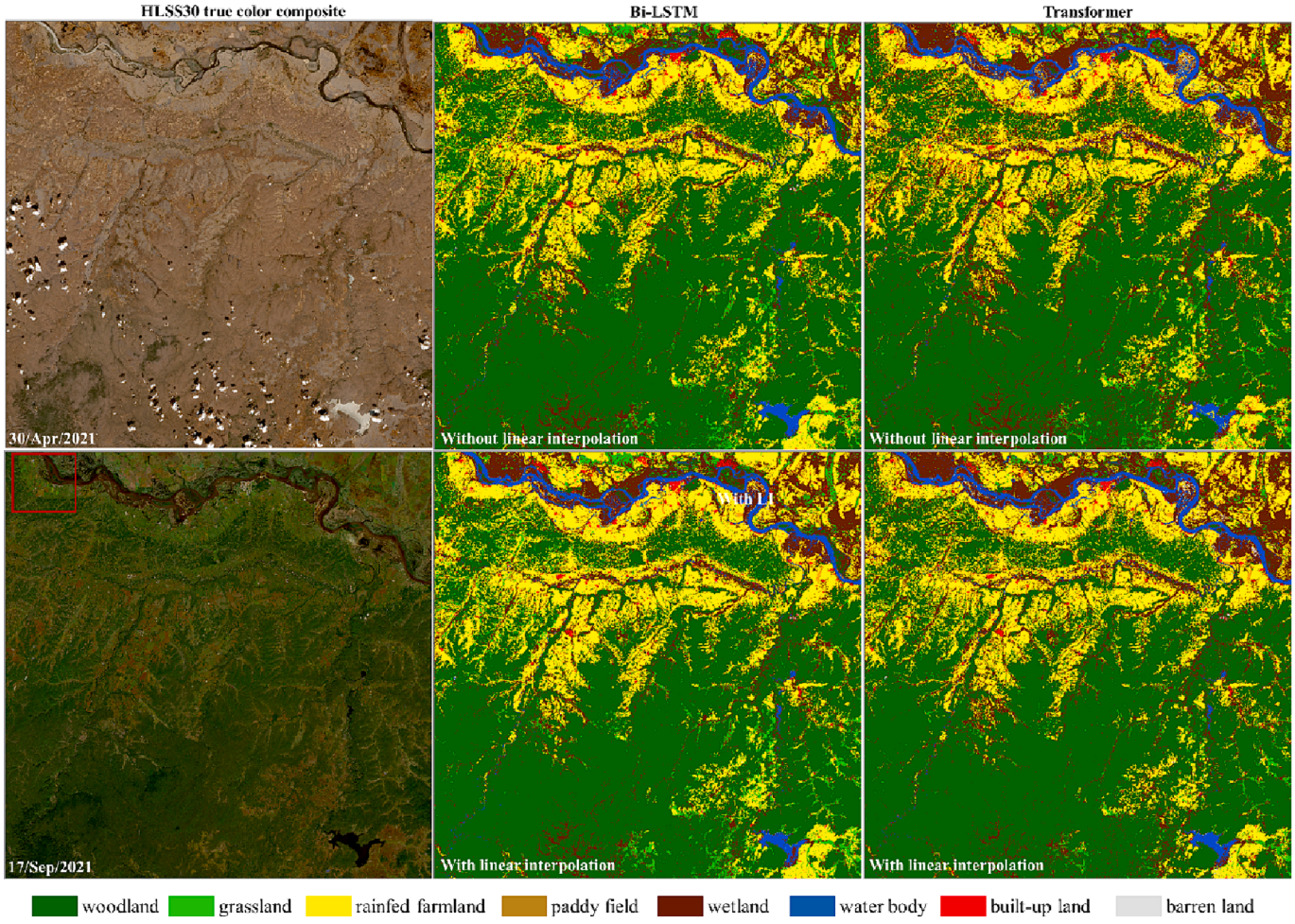

Fig. 11. The true color image (left column), and land cover classification maps derived from Bi-LSTM (middle column) and Transformer (right column) models for time series composites without and with linear interpolation for the tile 52UDV.

4.5. Computation time comparison

Fig. 16 shows the computation time for the classification of the tile 52UDV and 51TXL composites without and with linear interpolation for the Bi-LSTM and Transformer models. The computation time includes image I/O, composite, linear interpolation (if needed) and classification of composite with or without linear interpolation. It indicates more than 13 times computation time can be saved for the classification without linear interpolation compared to that with linear interpolation for both Bi-LSTM and Transformer models. The computation time of Transformer model is 18 % smaller than Bi-LSTM models to classify composite without linear interpolation. This is probably attributed to the fact that Transformer model can be run in parallel for each 16-day temporal period while Bi-LSTM hidden feature extraction for each time period depends on the output at the previous time period. By contrast, there is little computation time difference between Bi-LSTM and Transformer models to classify composites with linear interpolation, because linear interpolation occupies most of the computation time compared to composite and classification.

5. Discussion

5.1. Why linear interpolation does not help

The contribution of linear interpolation of missing values was found to be little for deep learning models that can handle missing data (i.e.,

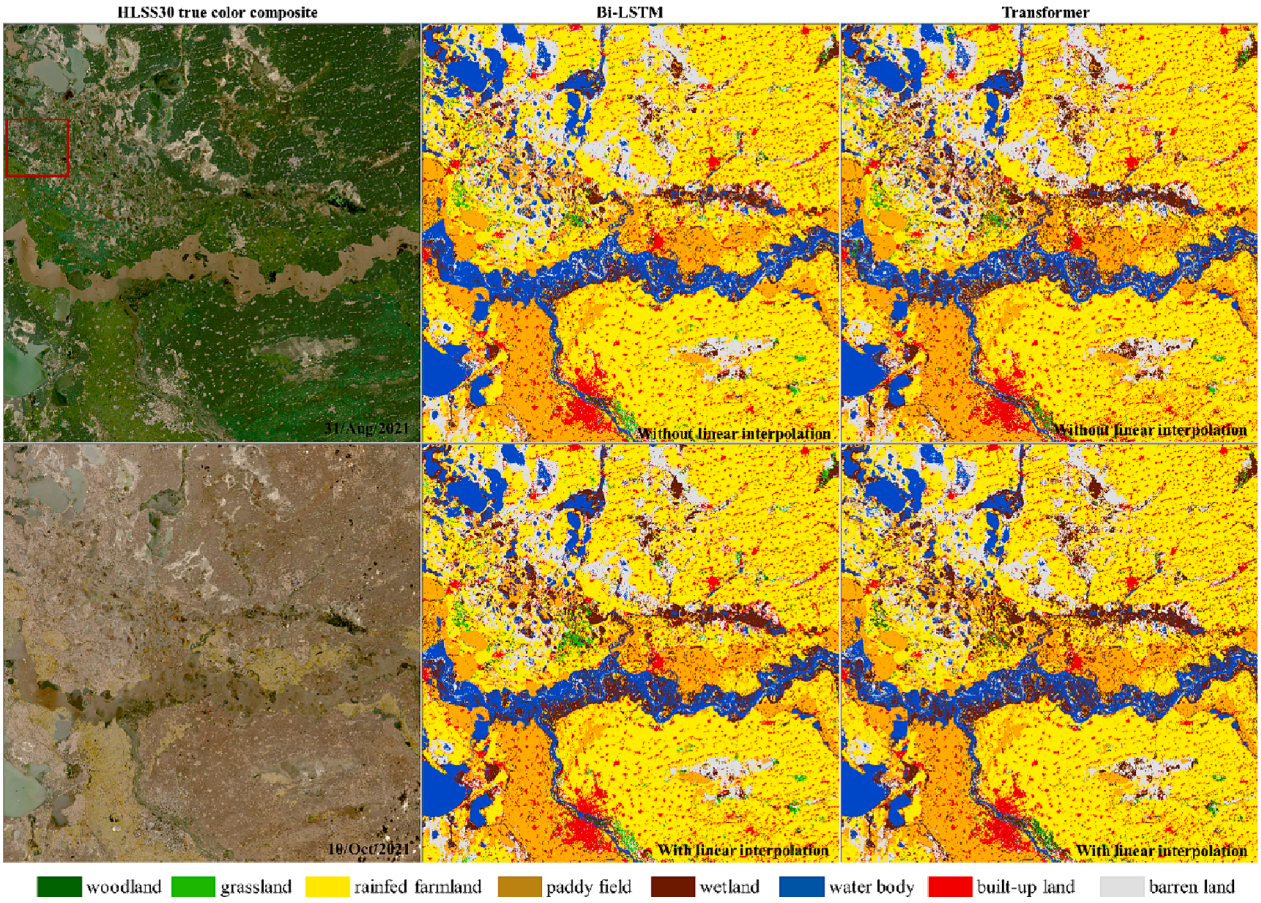

Fig. 12. The true color image (left column), and land cover classification maps derived from Bi-LSTM (middle column) and Transformer (right column) models for time series composites without and with linear interpolation for the tile 51TXL.

masking out missing values in the model). We compared land cover classifications of single pixel time series composites without and with linear interpolation using the Bi-LSTM and Transformer models using 17,963 30 m HLS time series images over 144 HLS tiles acquired over Amur River Basin of China from March to October in 2021. The comparisons show the land cover classification accuracies of composites without linear interpolation are generally comparable to those with linear interpolation for both the Bi-LSTM and Transformer models. The computation time for classification without linear interpolation can be 13 times less than the classification composites with linear interpolation. This indicates the time-consuming linear interpolation is not necessary for land cover mapping using Bi-LSTM and Transformer models.

The reason why linear interpolation does not help should lie in that linearly interpolated time series brings no additional information compared to the time series with missing values. We discussed this in two different cases (i) the linear interpolation reconstructs the time series well as the reflectance follows linear change, e.g., when the missing data period is short; and (ii) the linear interpolation fails to produce authentic observations as the linear change assumption is violated, e.g., when the missing data period is long. Fig. 17 simulated missing values by masking out certain time periods with good quality observations in four example pixel time series so that the interpolated values (orange) can be compared with the observed ones (blue dots with red squares). Case (i) was illustrated by the two example time series for the built-up land and forest and linear interpolation is implicitly performed in the LSTM and Transformer classifications. When the Bi-LSTM model meets the period with missing values, the model will skip the missing period (Section 3.2). We can also interpret the process as that the model does not skip but simply copies the feature vector from the previous period as the missing period feature vector in the forward LSTM and copies the feature vector from the successive period as the missing period feature vector in the backward LSTM. The combination of the two feature vectors (i.e., concatenation) acts like an interpolation. Similarly, when the Transformer model classifies the time series with missing values, the attention layer can adjust the weights for the feature vectors from previous and successive composite periods without missing values (Section 3.2) to produce similar output to the results of classifying the linear interpolated time series. Case (ii) was illustrated by the two time series for the double season (winter wheat and corn) crop and grassland types the linearly interpolated values are far from observed ones. In this case, linearly interpolated time series is incorrect for classification with linear interpolation, and similarly both LSTM and Transformer models cannot reliably classify the time series with missing values. Consequently, both the classifications with and without linear interpolation can have similar land cover classification accuracies despite that their classification results can be different.

5.2. *Sensitivity to other land cover classification legends*

The little contribution of linear interpolation to classification should be applicable to any land cover legends based on the discussion in Section 5.1 since linear interpolation brings little information to time series. To examine this, two different classification experiments with different legends to above studies were undertaken (i) using the same dataset as in Section 2 by excluding some classes exhibiting little seasonal variations in classification, and (ii) using two new sets of training and evaluation samples derived for crop type classification.

The experiment (i) used only five of the eight land cover types in Section 2.3, i.e., woodland, grassland, rainfed farmland, paddy field, and wetland types. The same five-fold cross validation (Section 3.3)

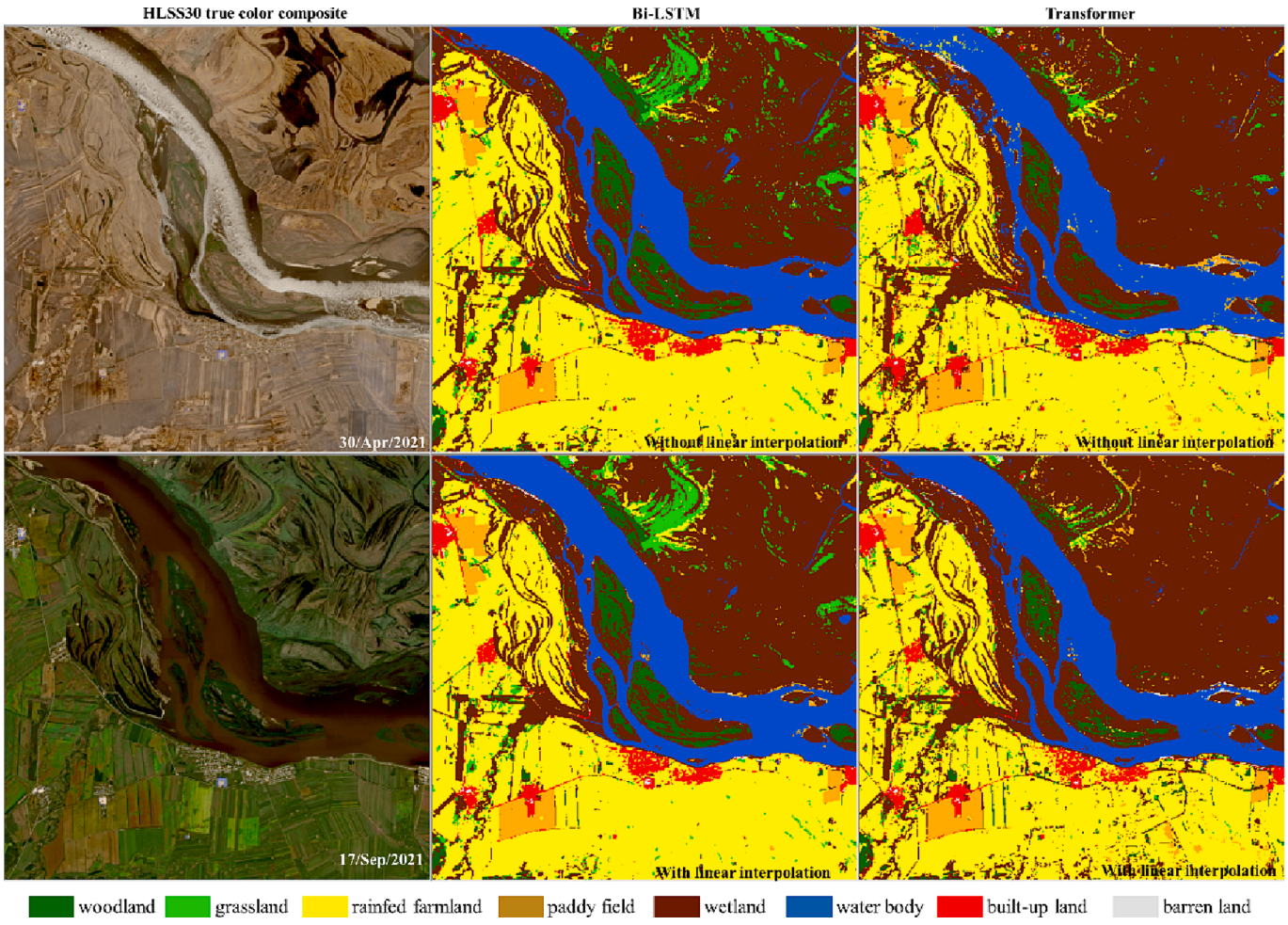

Fig. 13. Same as Fig. 11 but for the Fig. 11 red box area with the 500 × 500 30 m pixels. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

were used with each fold run for five times and only the Transformer models were used for fast computation. Table 7 shows the testing dataset overall accuracy for the Transformer model derived for composites without and with linear interpolation. The mean overall accuracy differences between classifications of composites without and with linear interpolation for the five type was < 0.5 % for Transformer models which equals to the corresponding differences of all eight types in Table 5. Table 8 displays F1 values of the testing sets for the Transformer models trained with all eight types and with five types. The F1 values for five land cover types have little difference (<0.003) between classifications of composites without and with linear interpolation.

The experiment (ii) focused on crop type classification given that crop type classification typically requires more temporal information than general land cover classification (Zhong et al., 2019; Xu et al., 2020; Wang et al., 2020; Li et al., 2023). The same five-fold cross validation (Section 3.3) was used with each fold run for five times and the Transformer model was used to classify the HLS 16-day time series composite with missing values and the linearly interpolated time series composite with no missing values over crop samples. We collected crop type training data from the existing land cover products which have been widely used for training data derivation (Knorn et al., 2009; Xian et al., 2009; Sexton et al., 2013; Jia et al., 2014; Wessels et al., 2016; Tran et al., 2022). We used the 30 m crop data layer (CDL) product which is an annual crop type for the U.S. at 30 m resolution with more than 100 crop types and is generated by the U.S. Department of Agriculture's National Agricultural Statistics Service (NASS) with the high classification accuracy of 85 % to 95 % for majority crops (Boryan et al., 2011). Two study areas were selected which are located in California (HLS tile T11SKA) and Oklahoma (T14SPF), respectively. We produced reliable crop type training samples for each study area using the method developed by our previous study (Tran et al., 2022). Specifically, the CDL is sampled every 20 pixels (600 m) to reduce spatial autocorrelation (Yang et al., 2003) and those samples with different land cover labels in eight adjacent pixels from the pixel label are filtered out. The California tile has significantly different fruit and nut crops samples including grapes, oranges, citrus, almonds, walnuts, and pistachios while winter wheat and soybeans are the two most common crops in the Oklahoma HLS tile (Table 9). Only land cover types with greater than 100 pixels were used for training the Transformer model, and the sample number of each land cover type was summarized in Table 9 for both tiles. Table 9 also shows the annual percentages of the samples with missing values for each land cover type, and both of two tiles had an average of nearly 40 % samples with missing values. The California and Oklahoma tiles had the most missing values for evergreen forest type and water body type, respectively.

Tables 10 and 11 show testing dataset overall accuracies for Transformer classification of time series composites without and with linear interpolation over tile T11SKA and T14SPF, respectively. The mean overall accuracy differences between classifications without and with linear interpolation was < 0.3 % for tile T11SKA and < 0.2 % for tile T14SPF. For tile T11SKA, the mean overall accuracies for Transformer classification of time series composite without linear interpolation ranged from 88.7 % to 88.9 % while those with linear interpolation ranged from 88.8 % to 89.0 %. In contrast, the mean overall accuracies

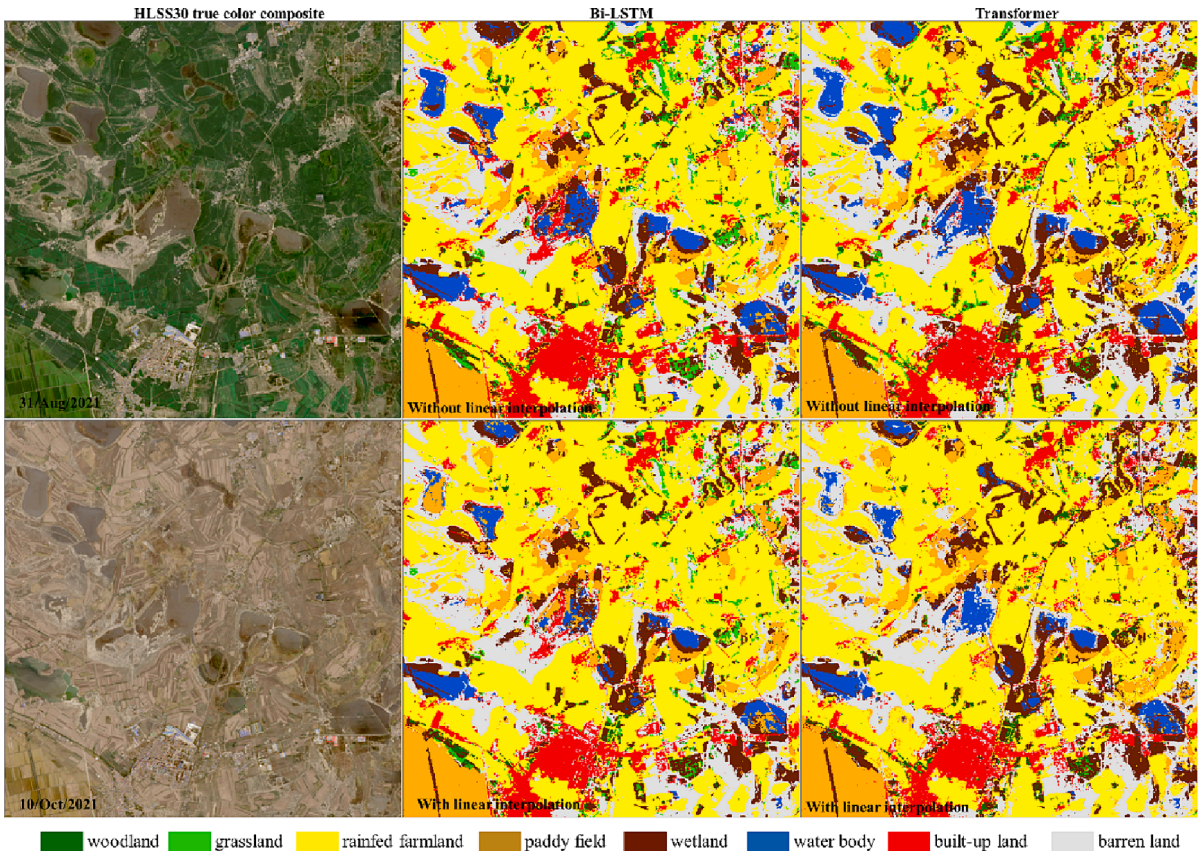

Fig. 14. Same as Fig. 12 but for the Fig. 12 red box area with the 500 × 500 30 m pixels. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

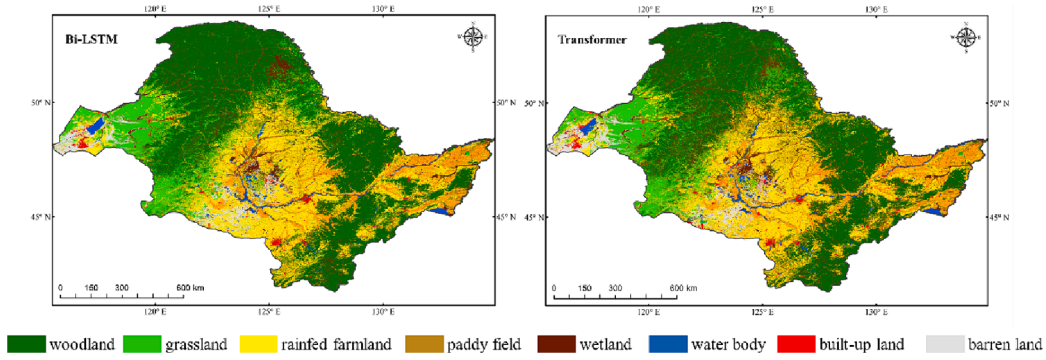

Fig. 15. Land cover classification maps derived from Bi-LSTM (left) and Transformer (right) models for image composites without linear interpolation for the study area (Amur River Basin).

for Transformer classification of time series composite without linear interpolation ranged from 95.5 % to 95.7 % for the tile T14SPF while those with linear interpolation ranged from 95.4 % to 95.6 %.

Table 12 and 13 shows the mean and standard deviation of the testing dataset F1 values for each land cover type using the Transformer models run for five times over tile T11SKA and T14SPF, respectively. The mean F1 values of seventeen land cover types have little difference for classification of composites without and with linear interpolation with < 0.006 except for relatively less Double crop Triticale/Corn type for the tile T11SKA, and < 0.008 except for relatively less Other Hay/Non Alfalfa type for the tile T14SPF. All mean F1 difference values between classifications of composites without and with linear interpolation are lower than the corresponding standard derivations. The results indicate there is no significant accuracy difference for classification of composites without and with linear interpolation for crop types.

We found the insignificant accuracy differences between classifications using composites with and without linear interpolation are also applicable to crop type classification although the crop type often has more complicated spectral change across a year due to the management practices, such as harvesting and tillage (Luo et al., 2023). This verified that linearly interpolated time series brings no additional information for crop type classification compared to the time series with missing values in Transformer classification which is consistent with the conclusions for general land cover in Section 5.1. In fact, a similar conclusion is drawn by (Zhao et al., 2021) using Sentinel-2 time series with missing values for crop type mapping using deep learning models such as LSTM and gate recurrent unit (GRU).

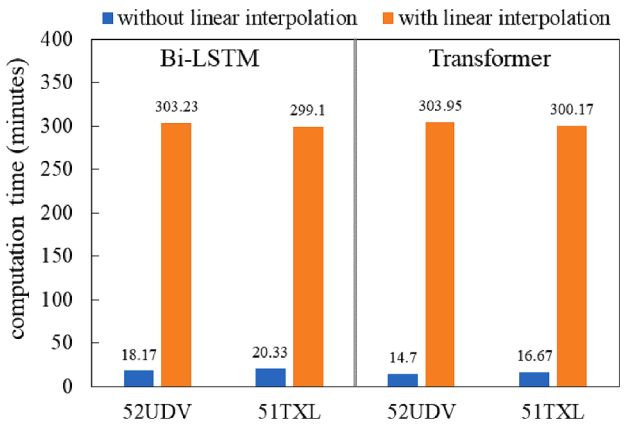

Fig. 16. The computation time (minutes) for the classification of the tile 52UDV and 51TXL composites without and with linear interpolation for the Bi-LSTM and Transformer models.

5.3. Sensitivity to number of good-quality composite in time series

In the China study area (Section 2.1) there is on average 15.35 % composite periods with missing values. However, it is unknown whether the linear interpolation can help in areas with more frequent cloud cover, e.g., in Great Lakes area or in Amazon (Asner, 2001; Whitcraft et al., 2015; Eberhardt et al., 2016; Egorov et al., 2019). To examine this, classification experiments were undertaken by randomly masking out 50 % of observations for each time series as clouds for the China study area. The 50 % random masking procedure was run five times to guarantee diversity in the observation temporal distributions in time series. Fig. 18 shows examples of original and randomly-masked time series satellite observations for eight different land cover samples. Using the randomly-masked time series, Transformer models are trained using data with and without linear interpolation to classify the eight land cover types. The mean overall classification accuracy differences derived from composites without and with linear interpolation was < 0.3 % in Table 14 which is consistent with the results using the original time series in Table 5. The F1 values for the Transformer models trained with original and randomly masked time series shows similar patterns where the mean F1 values difference were < 0.008 and 0.006, respectively (Table 15). Furthermore, all classification accuracy results using the original time series observations are relatively higher than that using the randomly masked time series. This is expected since the random masking of time series would possibly loss important temporal dynamics information (van der Wal et al., 2013).

To further examine this, we also undertook another classification experiment by randomly masking out different percentages of the annual 16-day composites for two agricultural study areas in Section 5.2. We only used samples without any missing values in the 14 compositing periods over California and Oklahoma tiles so that they can be used to simulate various percentages of composites periods being masked out as missing values. And there are 18,539 samples with 19 classes (grass/pasture, winter wheat, almonds, fallow/idle cropland, grapes, pistachios, soybeans, developed, double crop winter wheat/soybean, deciduous forest, corn, alfalfa, citrus, sorghum, tomatoes, walnuts, double crop winter wheat/corn, cotton and open water) by combination of the two tiles (Table 9). The random masking was run to make the percentage of the annual 16-day composites with good-quality observations ranging from 10 % to 100 % with step of 10 %. Fig. 19

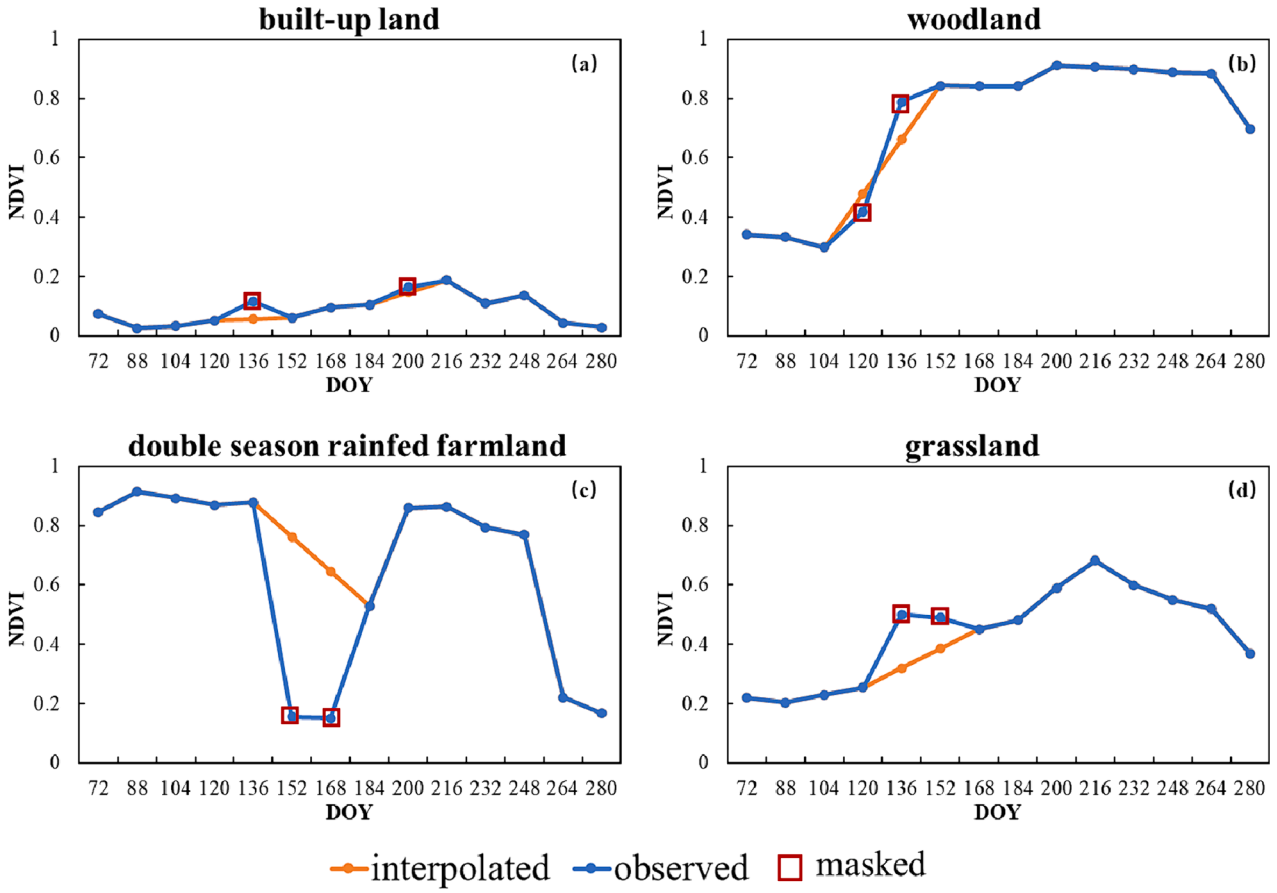

Fig. 17. Differences between interpolated NDVI (orange dots) and observed NDVI (blue dots with red squares) over four example pixels with different land cover types. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 7
Same as Table 5 but trained with all the samples excluding barren land, built-up, water body type.

Run	Linear interpolation	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean ± SD
#1	without	89.5	91.0	89.1	90.8	89.6	90.0 ± 0.9
	with	89.4	90.4	87.8	90.9	88.9	89.5 ± 1.2
#2	without	89.6	91.4	87.9	90.4	88.8	89.6 ± 1.3
	with	89.4	89.7	88.7	90.9	89.5	89.6 ± 0.8
#3	without	89.4	90.5	88.4	91.7	88.6	89.7 ± 1.3
	with	89.5	89.9	88.7	91.6	89.4	89.8 ± 1.1
#4	without	88.4	90.8	87.8	91.7	87.7	89.3 ± 1.8
	with	89.5	91.0	88.6	90.1	89.6	89.8 ± 0.9
#5	without	88.7	91.9	87.6	91.4	89.6	89.8 ± 1.8
	with	89.5	90.6	88.6	91.3	88.4	89.7 ± 1.2
Mean ± SD	without	89.1 ± 0.5	91.1 ± 0.6	88.2 ± 0.6	91.2 ± 0.6	88.9 ± 0.8	
	with	89.4 ± 0.1	90.3 ± 0.5	88.5 ± 0.4	91.0 ± 0.6	89.1 ± 0.5	

shows the overall accuracies for classifications without and with linear interpolation using 5-fold cross-validation as the function of percentages of the 16-day composites with good-quality observations. As the percentage of 16-day composites with good-quality observations increased, the accuracy first increased and then stabilized for both classifications. This is expected because more temporal dynamics information was captured with the increasing number of 16-day composites with good-quality observations. The overall accuracy differences between classifications without and with linear interpolation were almost negligible when the number of the 16-day composites with good-quality observations was higher than 5 (=14 16-day composites × 30 %). Their difference ranged from 1 % to 5 % with higher accuracy for the classification without linear interpolation when the number of the 16-day composites with good-quality observations was smaller than 5. This demonstrated linear interpolation probably produced unreliable time series when the number of 16-day composites with good-quality composites was small. The F1 values of each land cover type for classification without and with linear interpolation showed similar patterns (Fig. 20). There were almost no F1 values differences between classifications without and with linear interpolation for the relatively stable land cover types such as the developed, fallow/idle farmland and open water regardless of the number of annual 16-day composites with good-quality composites. This is expected since the time series of these types are stable and the linear interpolation will bring little contribution for classification.

5.4. Other non-linear interpolation methods

Due to the simplicity and wide usage of linear interpolation, this study only demonstrated the linear interpolation to reconstruct missing values. There are numerous non-linear interpolations to reconstruct missing values such as spatial-based methods of kriging interpolation (Rossi et al., 1994; Addink and Stein, 1999) and Neighborhood Similar

Table 8
F1 values comparison of the testing sets between all eight types and five types for the Transformer models. The mean and standard deviation of five times run are shown.

Type	With samples of eight types		With samples of five types	
	without linear interpolation	with linear interpolation	without linear interpolation	with linear interpolation
woodland	0.920 ± 0.007	0.921 ± 0.005	0.924 ± 0.004	0.924 ± 0.001
grassland	0.880 ± 0.005	0.883 ± 0.006	0.891 ± 0.005	0.889 ± 0.005
paddy field	0.927 ± 0.003	0.932 ± 0.002	0.931 ± 0.005	0.934 ± 0.002
rainfed farmland	0.876 ± 0.005	0.884 ± 0.006	0.889 ± 0.004	0.890 ± 0.003
wetland	0.831 ± 0.008	0.836 ± 0.004	0.860 ± 0.004	0.858 ± 0.004

Table 9
The numbers of derived training samples and the annual percentage of samples with missing values for each land cover type.

California tile				**Oklahoma tile**			
Land cover type	**Code**	**Samples**	**% of samples with missing values**	**Land cover type**	**Code**	**Samples**	**% of samples with missing values**
Grass/Pasture	176	2909	64.52	Grass/Pasture	176	7324	47.56
Almonds	75	2553	37.84	Winter Wheat	24	4422	23.34
Grapes	69	2002	47.75	Soybeans	5	1097	33.45
Developed	121	1857	70.38	Deciduous Forest	141	949	50.26
Fallow/Idle Cropland	61	1400	24.64	Dbl Crop WinWht/ Soybeans	26	697	29.84
Pistachios	204	1380	35.14	Sorghum	4	494	20.65
Citrus	72	802	49.13	Corn	1	425	28.24
Alfalfa	36	555	25.59	Open Water	83	403	71.46
Tomatoes	54	539	29.31	Developed	121	336	47.62
Walnuts	76	469	36.46	Other Hay/Non Alfalfa	37	100	55.00
Evergreen Forest	142	409	92.67				
Dbl Crop WinWht/ Corn	225	387	31.00				
Winter Wheat	24	368	19.84				
Cotton	2	340	31.76				
Corn	1	231	31.60				
Dbl Crop Triticale/ Corn	228	120	24.17				
Safflower	33	115	31.30				

Table 10
Transformer testing dataset overall accuracy (%) derived for composites without and with linear interpolation using 5-fold cross-validation over California tile (T11SKA).

Run	Linear interpolation	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean ± SD
#1	without	89.1	88.8	88.7	88.8	89.0	88.9 ± 0.2
	with	88.8	89.7	89.0	88.9	88.6	89.0 ± 0.4
#2	without	88.8	89.5	88.7	89.1	88.6	88.9 ± 0.4
	with	88.7	89.4	88.7	89.3	88.4	88.9 ± 0.4
#3	without	88.5	89.5	88.8	88.5	88.4	88.8 ± 0.4
	with	88.7	89.9	88.7	88.7	88.6	88.9 ± 0.6
#4	without	88.5	89.5	88.7	88.3	88.3	88.7 ± 0.5
	with	88.7	89.8	88.7	88.8	88.7	89.0 ± 0.5
#5	without	88.4	90.0	88.4	88.4	88.7	88.8 ± 0.7
	with	88.3	89.4	88.5	89.0	88.8	88.8 ± 0.5
Mean ± SD	without	88.7 ± 0.3	89.5 ± 0.4	88.7 ± 0.1	88.6 ± 0.3	88.6 ± 0.3	
	with	88.6 ± 0.2	89.7 ± 0.2	88.7 ± 0.2	88.9 ± 0.2	88.6 ± 0.1	

Table 11
Same as Table 10 but for Oklahoma tile (T14SPF).

Run	Linear interpolation	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean ± SD
#1	without	95.8	95.3	95.6	95.1	95.6	95.5 ± 0.3
	with	95.3	95.9	95.9	95.0	95.6	95.5 ± 0.4
#2	without	95.9	95.6	96.1	95	95.9	95.7 ± 0.4
	with	95.6	95.6	96.2	94.7	95.8	95.6 ± 0.5
#3	without	95.4	95.6	95.8	95.1	95.8	95.6 ± 0.3
	with	95.3	95.2	95.9	95	95.6	95.4 ± 0.4
#4	without	95.4	95.6	96.1	95.0	95.8	95.6 ± 0.4
	with	95.7	95.5	96.0	95.0	95.6	95.5 ± 0.4
#5	without	95.9	95.6	96.3	95.0	95.8	95.7 ± 0.5
	with	95.7	95.5	96.2	95.0	95.8	95.6 ± 0.4
Mean ± SD	without	95.7 ± 0.3	95.5 ± 0.1	96.0 ± 0.3	95.0 ± 0.1	95.8 ± 0.1	
	with	95.5 ± 0.2	95.5 ± 0.3	96.0 ± 0.2	94.9 ± 0.1	95.7 ± 0.1	

Pixel Interpolator method (Chen et al., 2011), spline interpolation (Tang et al., 2021), spatio-temporal hybrid methods of window regression (WR) (Brooks et al., 2018) and Spectral-Angle-Mapper-based Spatio-Temporal Similarity (SAMSTS) (Yan and Roy, 2018). Although these methods are capable of capturing more temporal and/or spatial variations and producing more accurate results compared to linear interpolation, they tend to be more sophisticated and computationally intensive. For example, kriging interpolation requires incorporation of spatial correlation, estimation of variogram parameters, and calculation of optimal weights (Meng et al., 2013). Neighborhood Similar Pixel Interpolator method introduces additional complexity through the use of local polynomial fitting, adaptive neighborhood selection,

Table 12
F1 values of the testing sets for each land cover type for the Transformer models over California tile (T11SKA). The mean and standard deviation of five times run are shown.

Type	Transformer	
	without linear interpolation	with linear interpolation
Grass/Pasture	0.964 ± 0.002	0.965 ± 0.001
Almonds	0.860 ± 0.003	0.862 ± 0.005
Grapes	0.828 ± 0.002	0.827 ± 0.003
Developed	0.936 ± 0.002	0.938 ± 0.002
Fallow/Idle Cropland	0.852 ± 0.003	0.854 ± 0.003
Pistachios	0.887 ± 0.003	0.889 ± 0.004
Citrus	0.857 ± 0.005	0.857 ± 0.007
Alfalfa	0.933 ± 0.003	0.931 ± 0.002
Tomatoes	0.948 ± 0.003	0.949 ± 0.003
Walnuts	0.853 ± 0.004	0.854 ± 0.002
Evergreen Forest	0.972 ± 0.002	0.974 ± 0.005
Dbl WinWht/Corn	0.706 ± 0.009	0.711 ± 0.006
Winter Wheat	0.675 ± 0.010	0.678 ± 0.008
Cotton	0.922 ± 0.006	0.922 ± 0.005
Corn	0.763 ± 0.004	0.769 ± 0.011
Dbl Triticale/Corn	0.147 ± 0.024	0.170 ± 0.029
Safflower	0.947 ± 0.007	0.941 ± 0.011

Table 13
Same as Table 12 but for Oklahoma tile (T14SPF).

Type	Transformer	
	without linear interpolation	with linear interpolation
Grass/Pasture	0.974 ± 0.001	0.974 ± 0.002
Winter Wheat	0.970 ± 0.001	0.970 ± 0.001
Soybeans	0.904 ± 0.003	0.901 ± 0.003
Deciduous Forest	0.939 ± 0.002	0.941 ± 0.003
Dbl Crop WinWht/ Soybeans	0.871 ± 0.002	0.870 ± 0.004
Sorghum	0.921 ± 0.008	0.913 ± 0.006
Corn	0.967 ± 0.004	0.968 ± 0.005
Open Water	0.985 ± 0.003	0.987 ± 0.001
Developed	0.904 ± 0.010	0.901 ± 0.004
Other Hay/Non Alfalfa	0.484 ± 0.009	0.461 ± 0.016

consideration of nonlinear relationships, and treatment of edge effects (Chen et al., 2011). Spatio-temporal hybrid methods need to consider both spatial and temporal dimensions, formulate hybrid model, handle non-stationarity, and utilize complex regression techniques (Brooks et al., 2018). SAMSTS requires utilization of spectral angle mapping, consideration of spatio-temporal similarity, feature extraction and dimensionality reduction, and parameter tuning (Yan and Roy, 2018). More recently, the research group developed a self-supervised learning method for interpolation where the time series pattern was learned by randomly masking out a few good-quality observations in a time series and training a Transformer model to predict the masked ones (Liu et al., 2024). These methods are seldom used for large region land cover classification due to their complication to use. Future studies to compare their contribution to land cover classification with deep learning models are encouraged. In addition to interpolation, some image spatiotemporal fusing strategies also have been proposed to generate fine temporal resolution images by blending MODIS and Landsat images, such as the widely-used Spatial and Temporal Adaptive Reflectance Fusion Model (STARFM) (Gao et al., 2006), the Enhanced STARFM (ESTARFM) (Zhu et al., 2010) and Flexible Spatiotemporal DAta Fusion (FSDAF) (Zhu et al., 2016). A few studies have used fused data for land cover classification and change detection (Jia et al., 2014; Li et al., 2017; Guo et al., 2020; Pi et al., 2023). Similar to linear interpolation, the contribution of these fusion approaches to time series based land cover classification using Bi-LSTM and Transformer models could be investigated in future research. It should be noted that fusion methods are different from

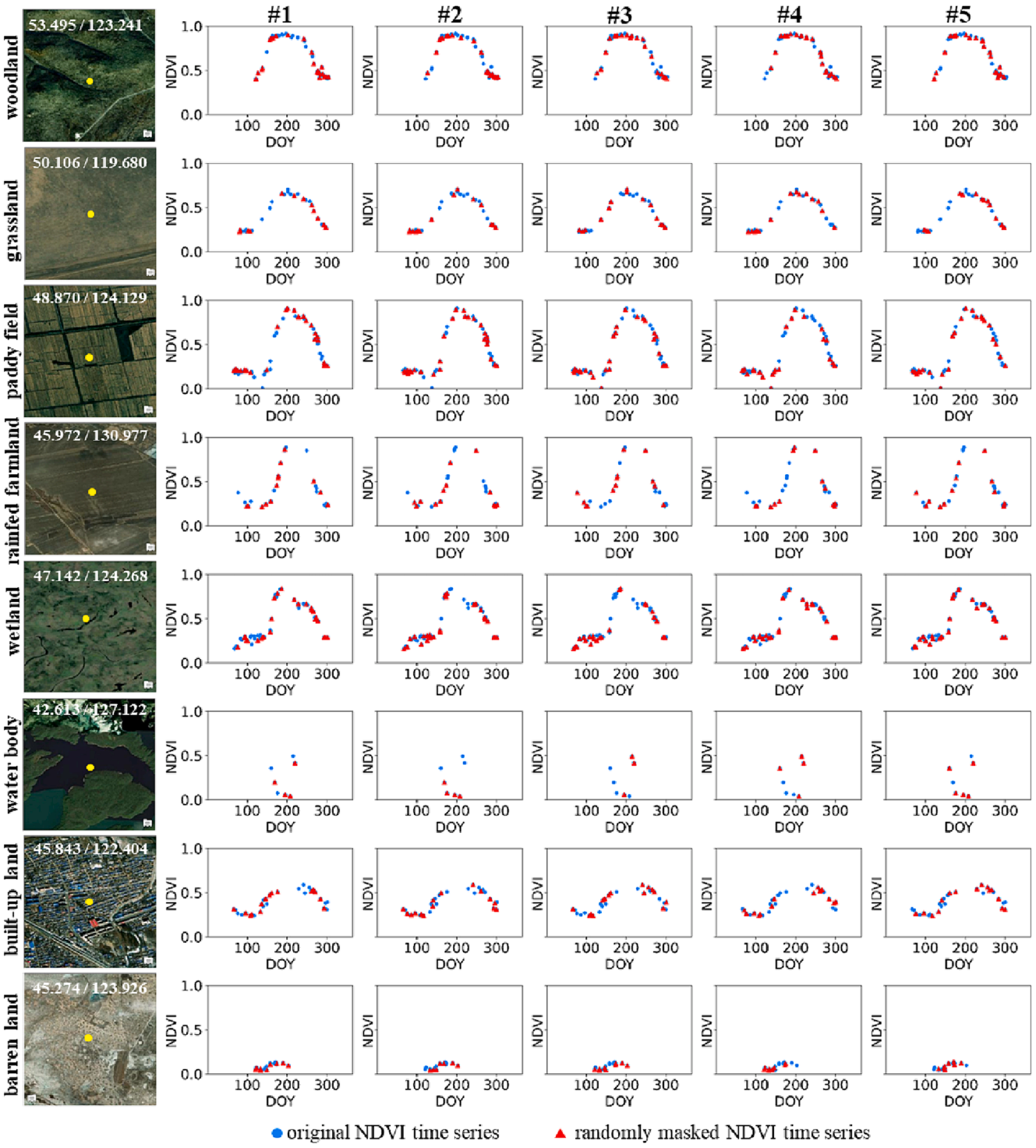

Fig. 18. Example pixel NDVI time series with the original (in blue) and 50% randomly masked time series (in red) for eight samples (First column show the google map image (with latitude and longitude) of eight different land cover samples, column 2–6 show five times random masking results). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

interpolation as they used additional coarse resolution data. Consequently, experiments should be carefully designed to reflect that the classification using data before and after fusion should be fair.

6. Conclusion

Satellite image time series data often have missing values due to the contamination of clouds, cloud shadows and snow. Numerous studies used linear interpolation to reconstruct missing values for land cover classification models, even for the RNN and Transformer structures that were designed to handle time series data with missing values. However, little is known about whether the time-consuming linear interpolation procedure contributes to the land cover classification for these deep learning models.

This study compared the land cover classification accuracy and efficiency for the time series composites without linear interpolation (with

Table 14
Same as Table 5 but trained with 50% randomly masked observations in time series of all eight types samples.

Sampling times	Linear interpolation	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean ± SD
#1	without	88.6	90.5	88.5	89.7	88.3	89.1 ± 0.9
	with	87.9	90.4	89.0	89.2	88.9	89.1 ± 0.9
#2	without	88.8	90.4	88.1	89.9	88.7	89.2 ± 0.9
	with	88.8	90.7	88.9	90.3	88.2	89.4 ± 1.1
#3	without	89.1	90.3	88.0	89.7	88.5	89.1 ± 0.9
	with	88.2	90.0	89.4	89.3	88.0	89.0 ± 0.8
#4	without	89.4	90.4	89.2	89.6	88.1	89.3 ± 0.8
	with	88.7	90.3	89.1	89.8	89.4	89.4 ± 0.6
#5	without	89.0	90.4	87.6	90.5	88.3	89.2 ± 1.3
	with	89.0	90.1	88.8	90.4	88.9	89.5 ± 0.7
Mean ± SD	without	89.0 ± 0.3	90.4 ± 0.1	88.3 ± 0.6	89.9 ± 0.4	88.4 ± 0.2	
	with	88.5 ± 0.5	90.3 ± 0.3	89.0 ± 0.2	89.8 ± 0.5	88.7 ± 0.6	

Table 15
F1 values comparison of the testing sets for the Transformer models trained with original and 50% randomly masked observations in time series of all eight types samples.

Type	With original time series of eight classes		With randomly masked time series of eight classes	
	without linear interpolation	with linear interpolation	without linear interpolation	with linear interpolation
woodland	0.920 ± 0.007	0.921 ± 0.005	0.907 ± 0.007	0.907 ± 0.004
grassland	0.880 ± 0.005	0.883 ± 0.006	0.870 ± 0.007	0.872 ± 0.005
paddy field	0.927 ± 0.003	0.932 ± 0.002	0.923 ± 0.005	0.927 ± 0.005
rainfed farmland	0.876 ± 0.005	0.884 ± 0.006	0.866 ± 0.007	0.872 ± 0.004
wetland	0.831 ± 0.008	0.836 ± 0.004	0.817 ± 0.007	0.815 ± 0.009
water body	0.940 ± 0.005	0.941 ± 0.003	0.938 ± 0.002	0.939 ± 0.006
built-up land	0.962 ± 0.002	0.963 ± 0.002	0.956 ± 0.003	0.953 ± 0.003
barren land	0.908 ± 0.001	0.908 ± 0.003	0.901 ± 0.006	0.897 ± 0.006

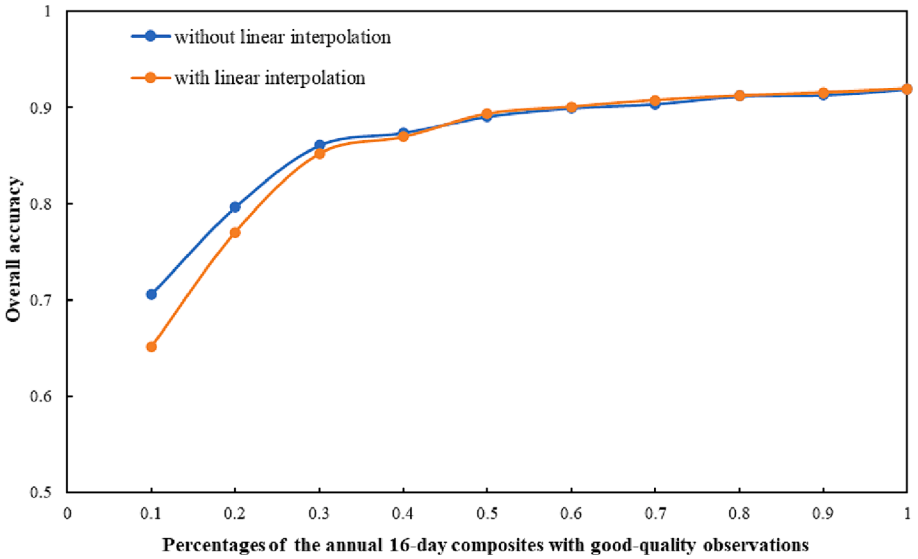

Fig. 19. The overall accuracies for classifications without and with linear interpolation using 5-fold cross-validation over California and Oklahoma tile as a function of the number of the 16-day composites with good-quality observations. *N* is the number of samples.

missing values) and the linearly interpolated time series composite (with no missing values) by taking advantage of the capability of handling time series data with missing values using Bi-LSTM and Transformer deep learning models. Each model was run for five times using 7755 samples with eight land cover types visually-interpreted with Landsat images time series from GEE. The results suggest that the classification accuracies were comparable between composites without and with linear interpolation, while the classification computation time without linear interpolation is nearly 13 times less than that with linear interpolation. The findings of this study indicate the time-consuming linear interpolation can be discarded for the Bi-LSTM and attention based models for single pixel based time series large region land cover classification. The models and comparison procedures developed in this study can be extended to investigate the necessity of other interpolation and fusing strategies for land cover classification. The training data (https://doi.org/10.5281/zenodo.7944466) and developed codes (https://github.com/chexianghong/Bi-LSTM-and-Transformer-classification-codes) are made publicly available for users to develop more advanced models or assess other interpolation methods.

This study implemented land cover classification using pixel-based multi-variables time series in the Bi-LSTM and Transformer models, and the spatial features are neglected which is crucial to land cover classifications especially for anthropogenic land cover such as cropland and built-up land covers with special textural features (Zhong et al.,

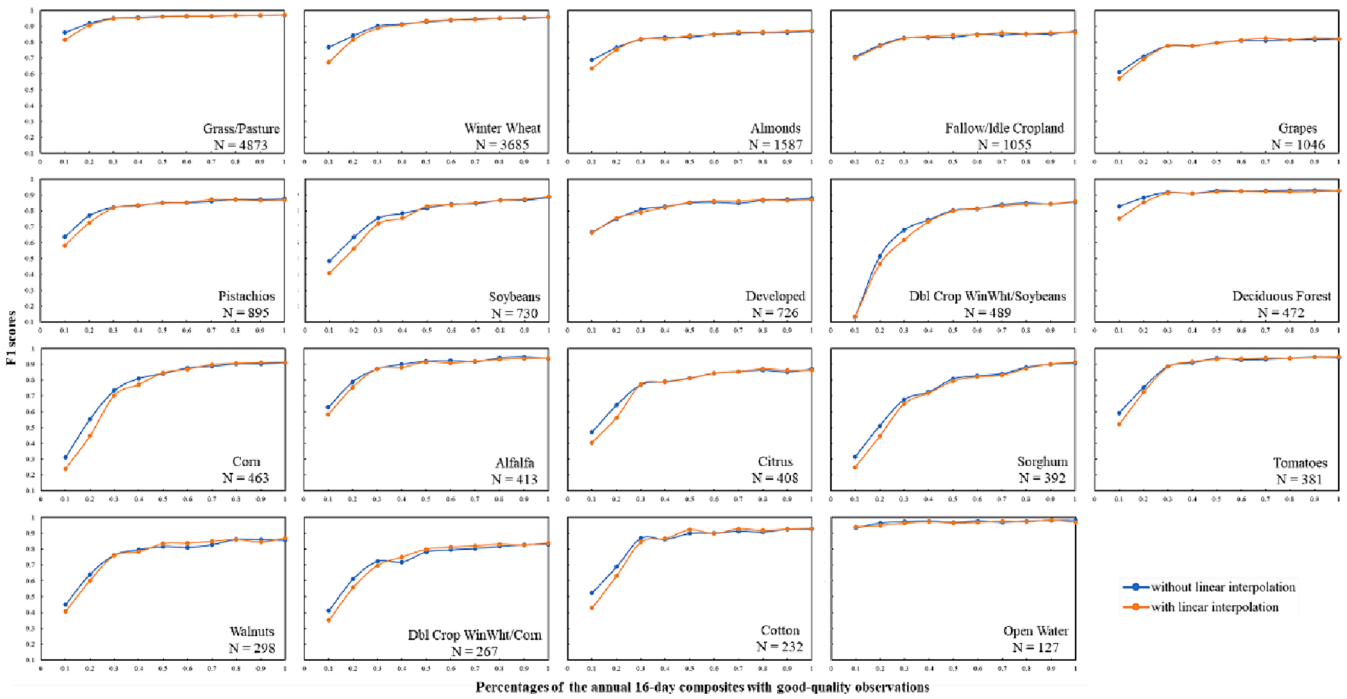

Fig. 20. F1 scores of each land cover for classifications without and with linear interpolation using 5-fold cross-validation over California and Oklahoma tile as a function of the number of the 16-day composites with good-quality observations. *N* is the number of samples.

2019). Future research comprehensively using temporal, spectral and spatial information for land cover classification is needed to improve the classification accuracy by combing convolution, recurrent network (Di et al., 2019) or spatial encoders to extract feature from parcels/objects (Garnot et al., 2020; Nyborg et al., 2022).

CRediT authorship contribution statement

Xianghong Che: Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Hankui K. Zhang:** Conceptualization, Formal analysis, Investigation, Software, Supervision, Writing – review & editing. **Zhongbin B. Li:** Funding acquisition, Writing – review & editing. **Yong Wang:** Writing – review & editing. **Qing Sun:** Writing – review & editing. **Dong Luo:** Conceptualization, Writing – review & editing. **Hao Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.isprsjprs.2024.04.021.

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